

EVS37

COEX, Korea, April 23-26, 2024

Quantifying the uncertainty of **EV** charging with probabilistic load forecasting

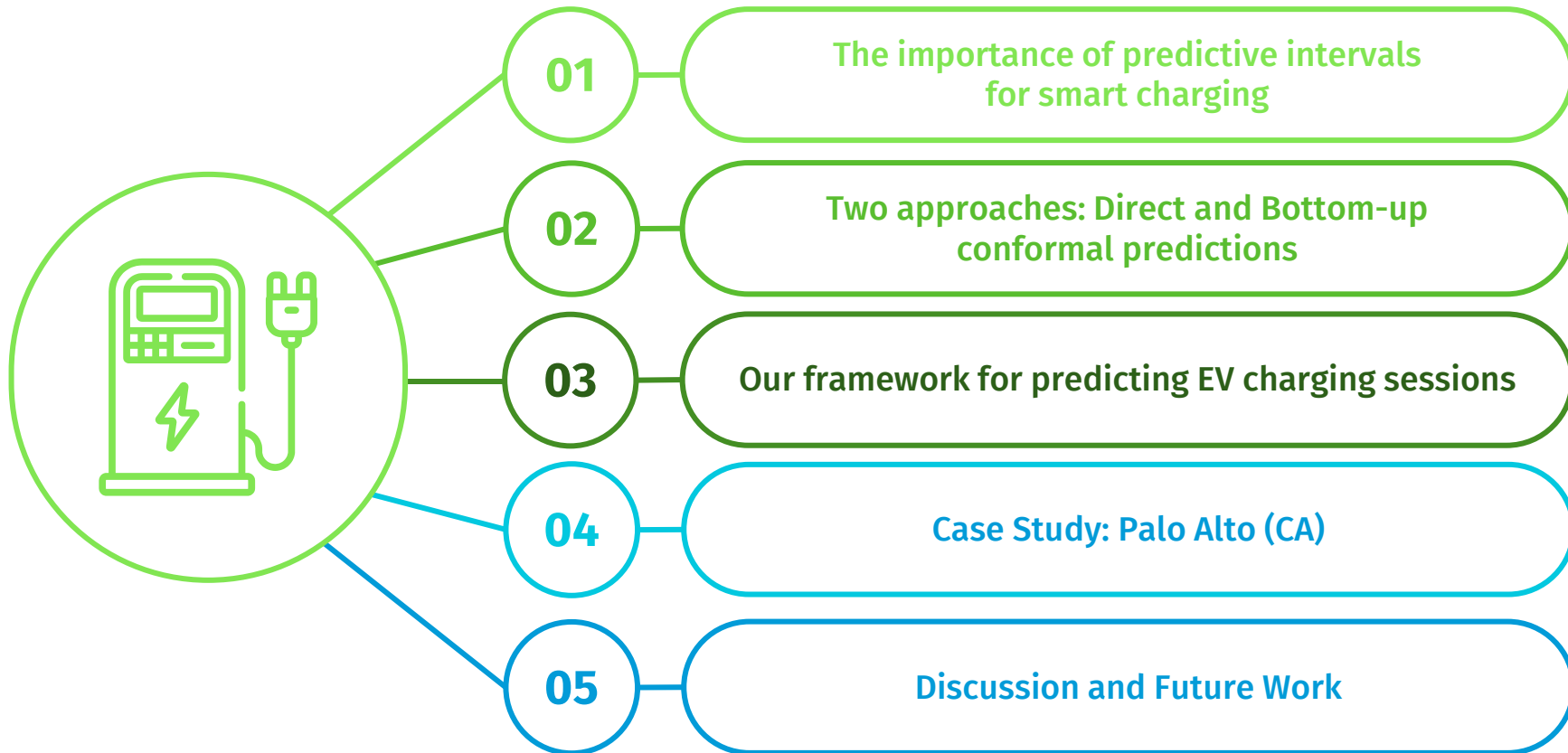
Yvenn Amara-Ouali^{1,2}, Bachir Hamrouche¹, Guillaume Principato^{1,2,3}, Yannig Goude^{1,2}

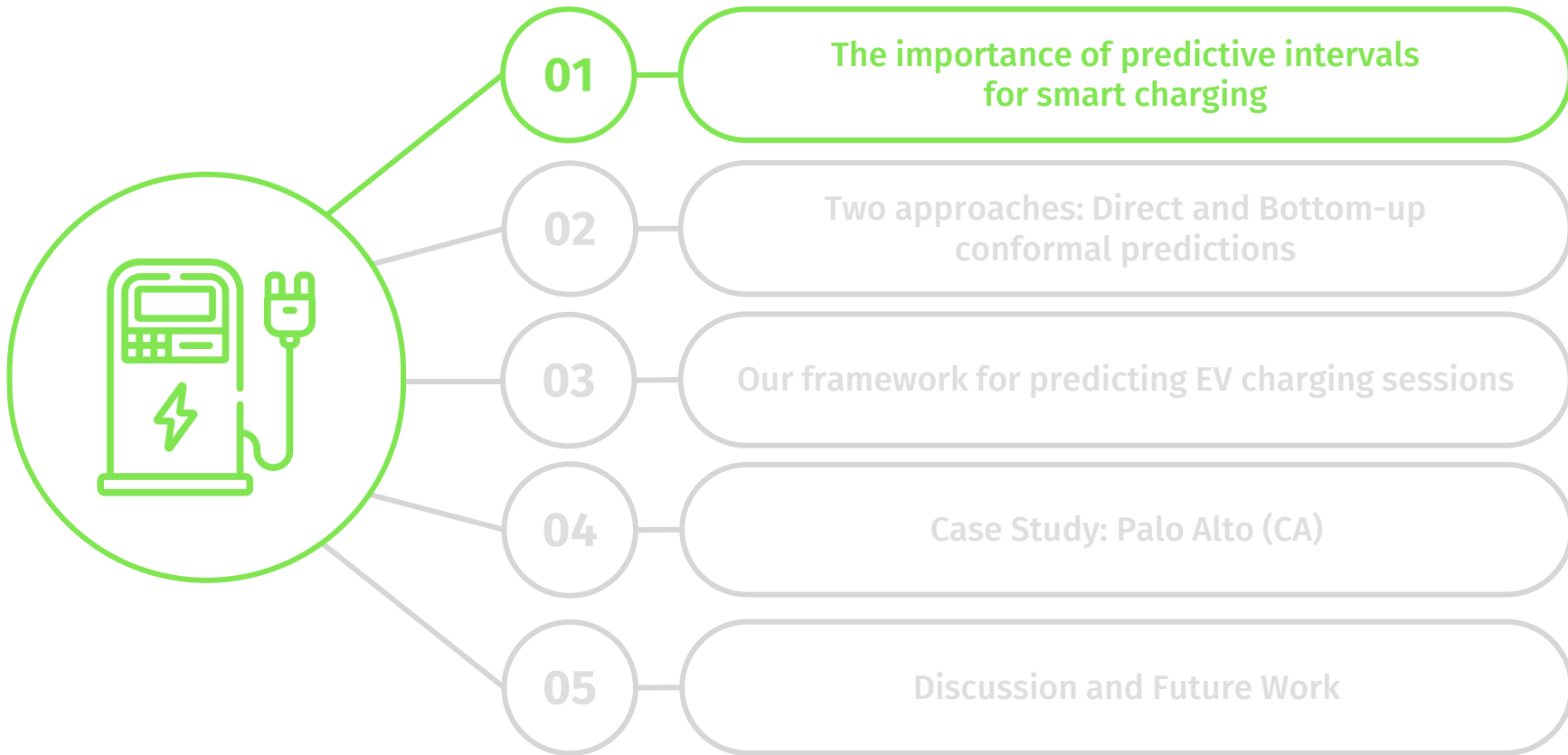
¹EDF R&D, Palaiseau, FRANCE

²University Paris-Saclay, LMO, Orsay, FRANCE

³Inria, CELESTE, Orsay, FRANCE

yvenn.amara-ouali@edf.fr

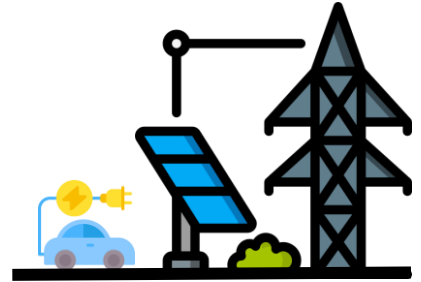




The importance of understanding EV charging behaviours

Situation

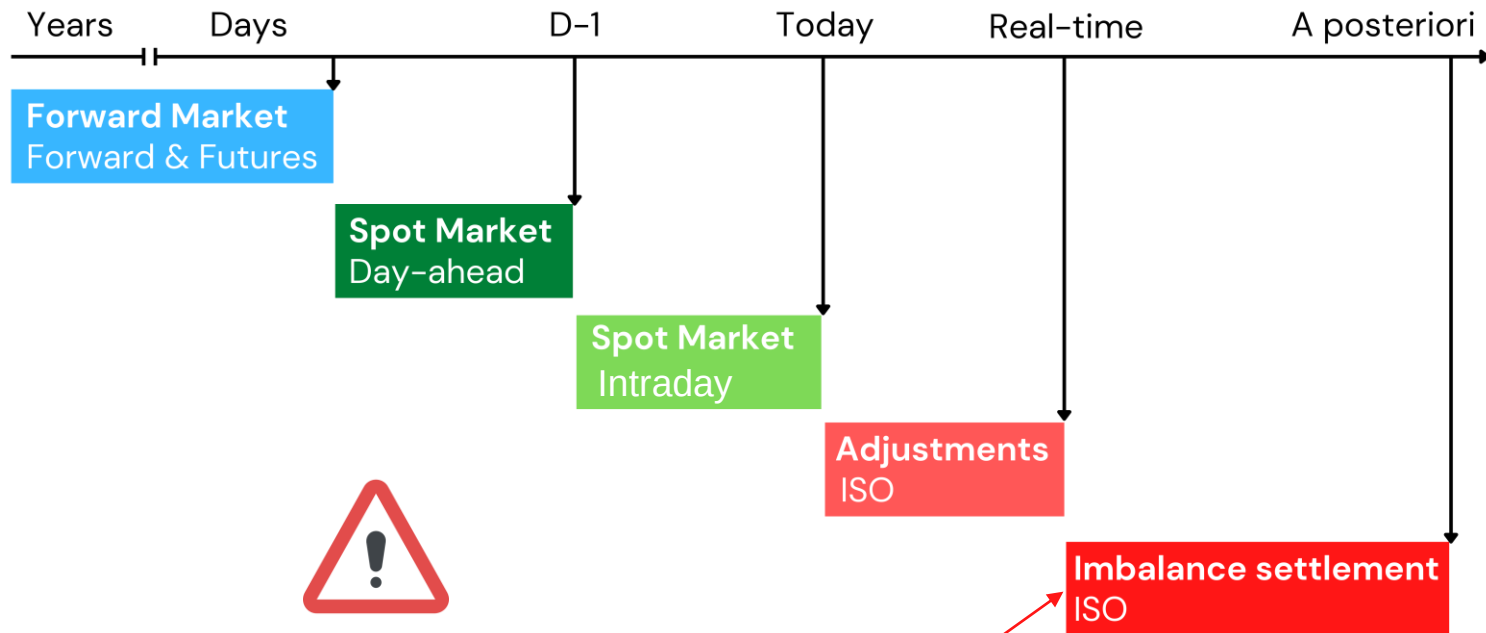
- In a world where resources are scarce, the goal is not to increase the number of power plants or electricity sources
- Instead, we want to better utilise what we already have by anticipating the electricity demand



Problematic

Producing **accurate forecasts** of EV charging demand
to enable **resilience** for the grid

Electricity markets simplified



Forecasting errors are penalised by the ISO

EV transactions at charging points

Sessions

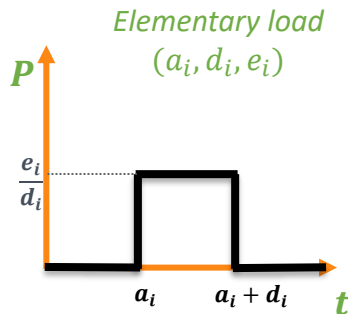
- The data often comes as a history of charging sessions
- Charging session = charging event for one vehicle

$$\text{Session } i = (a_i, d_i, p_i, e_i)$$

- a_i : arrival time
- d_i : charge duration
- p_i : parking duration
- e_i : energy demand

Elementary load profiles

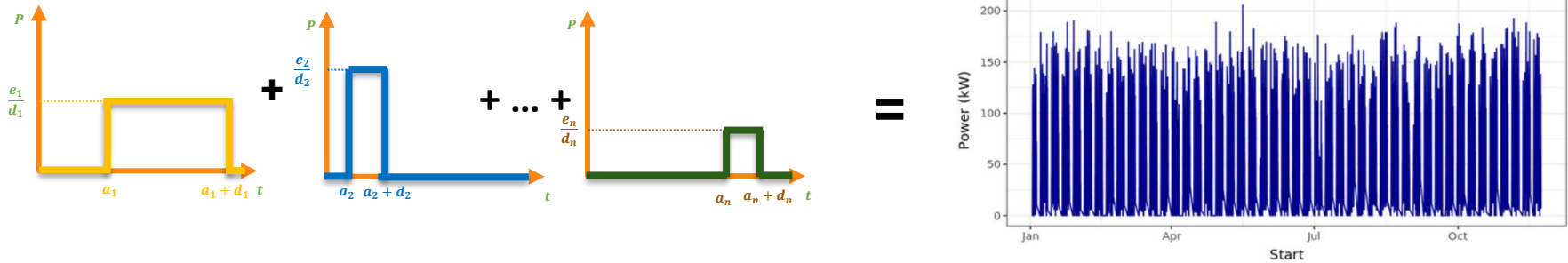
- For each session, an elementary load curve can be reconstructed from the session variables



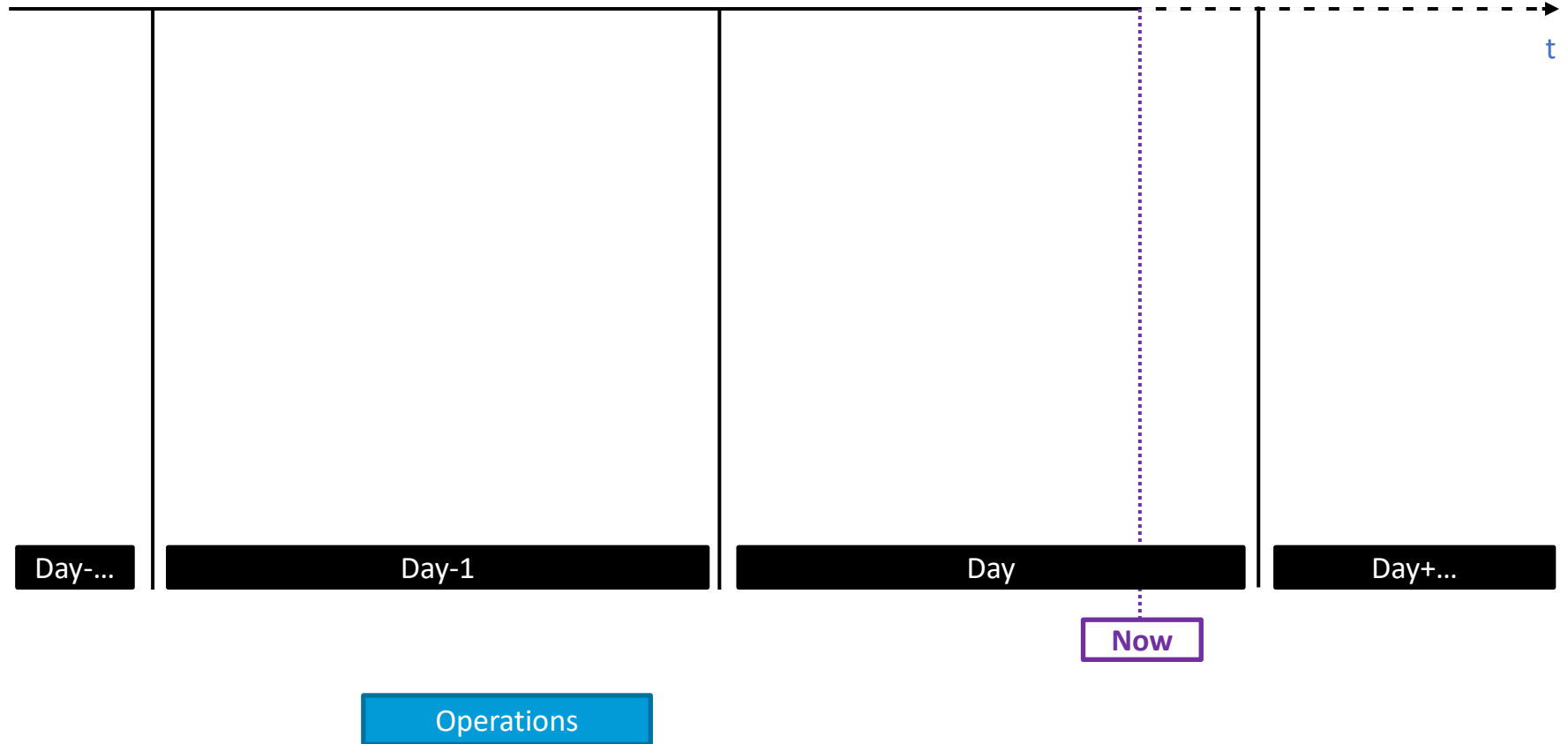
Profile reconstruction

Global load

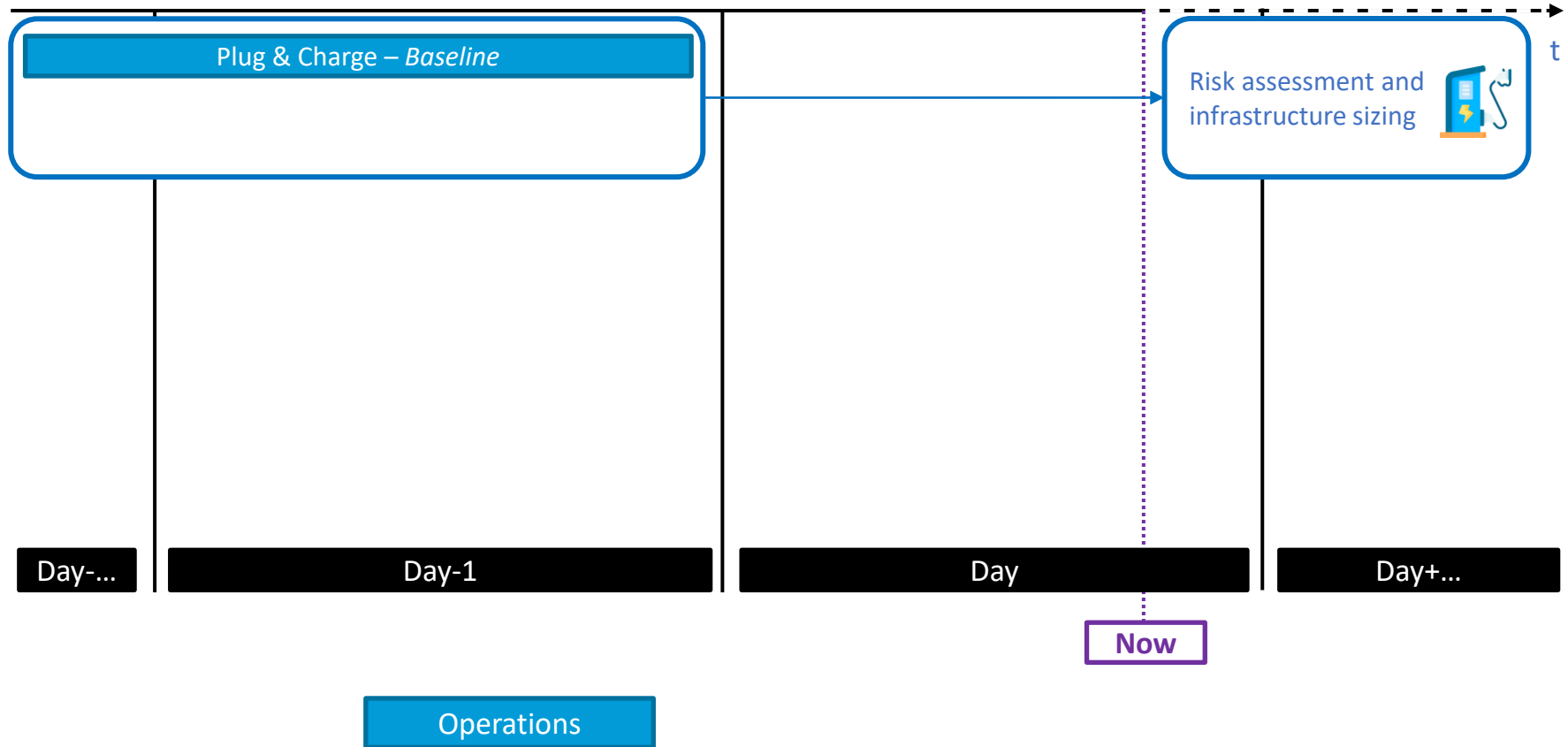
- Adding the elementary profiles leads to global profiles of interest at the scope of a station or dataset



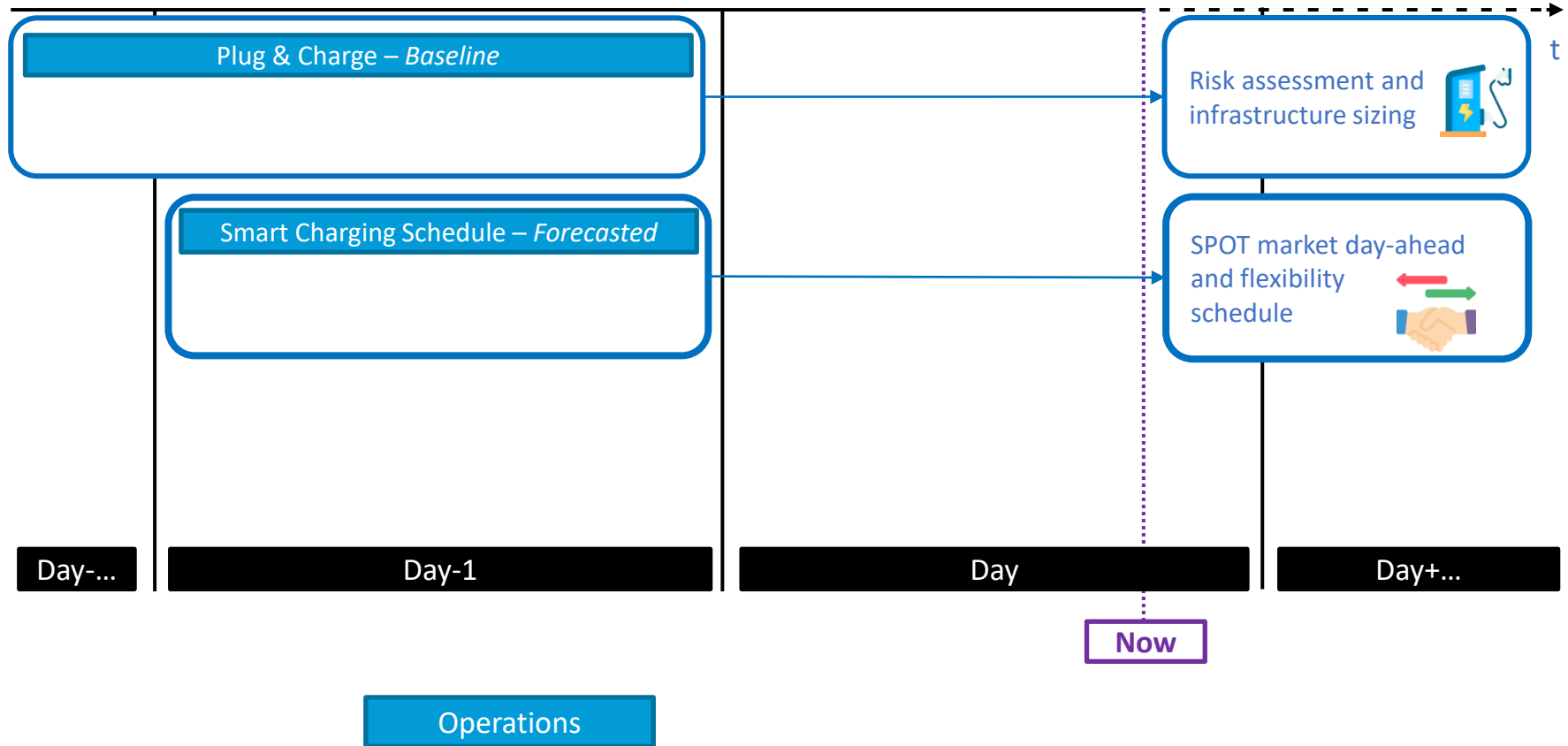
Operations



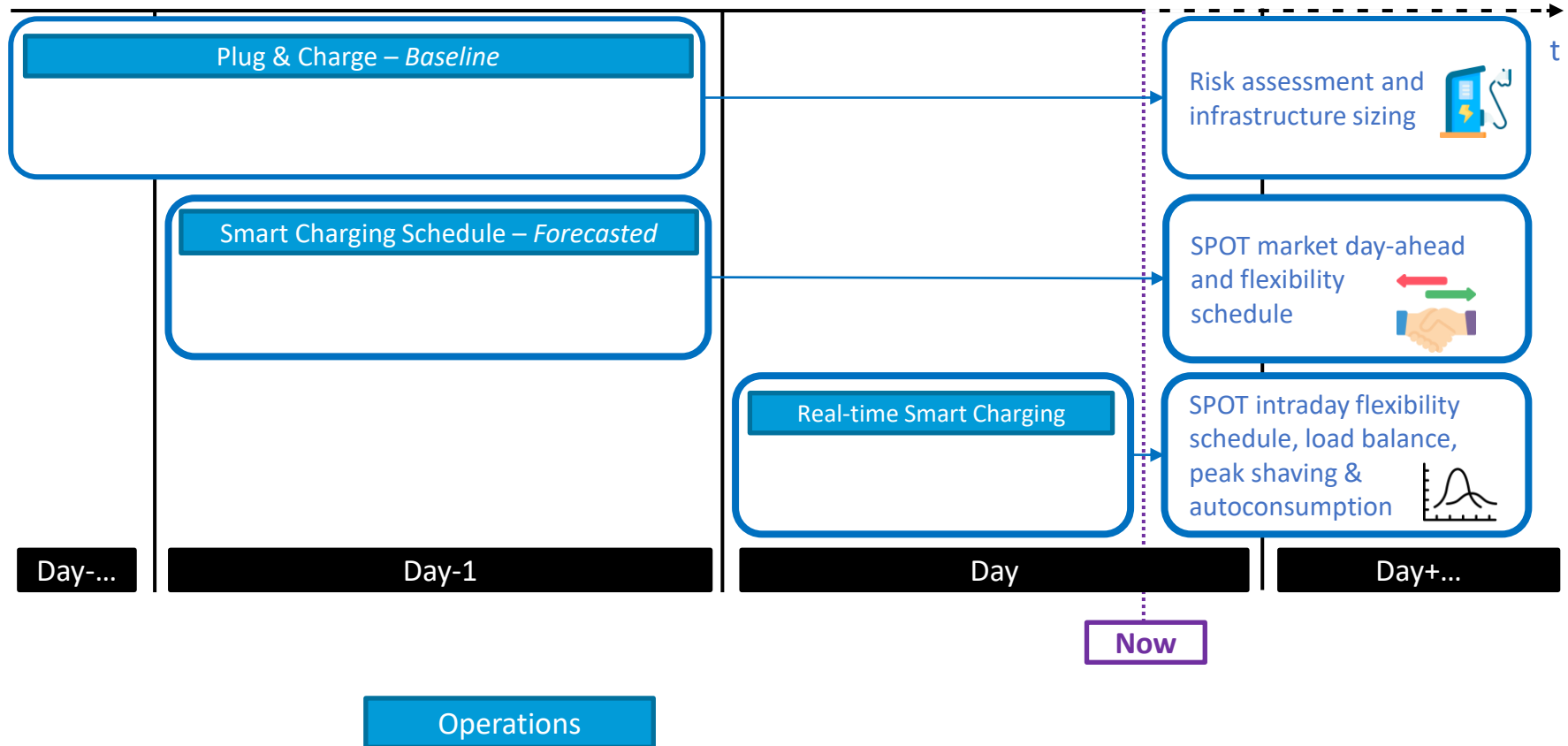
Operations



Operations



Operations



Building up on previous work

Main Results

- Automatic framework for assessing the performances of a wide variety of models across all datasets
- Direct approaches perform better than bottom-up approaches overall
- Bottom-up approaches outperform direct approaches on residential data

Limitations

- Exogenous information such as traffic and weather not considered

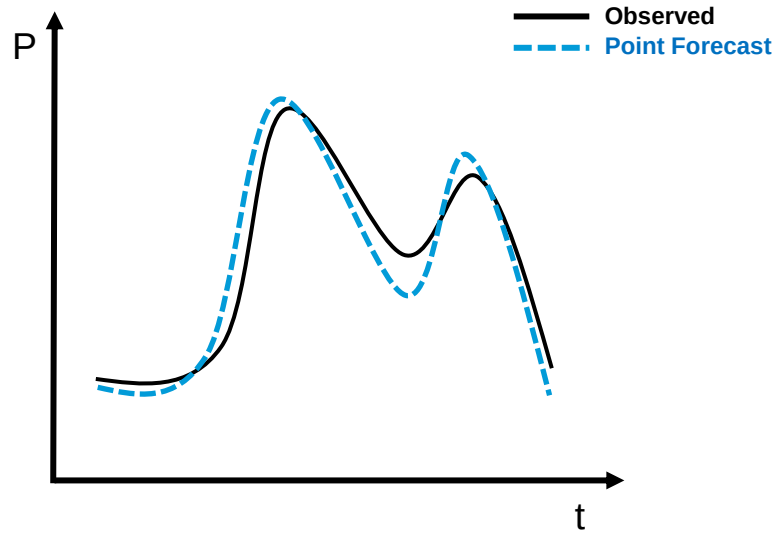
Perspectives

- Using the hierarchical structure of the data to improve bottom-up forecasts
- **Probabilistic forecasts of the EV charging demand**
- Improving the modelling performance on different datasets altogether (e.g., transfer learning)

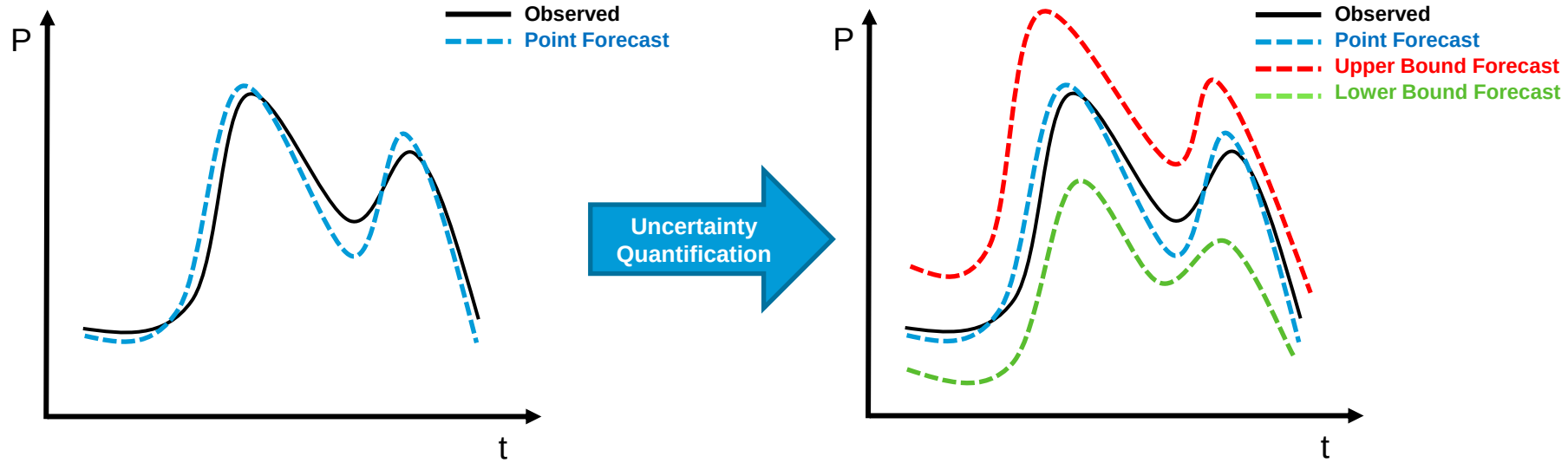
Publication

Amara-Ouali, Y., Goude, Y., Hamrouche, B., & Bishara, M. (2022). A benchmark of electric vehicle load and occupancy models for day-ahead forecasting on open charging session data. In *Proceedings of the Thirteenth ACM International Conference on Future Energy Systems* (pp. 193-207).

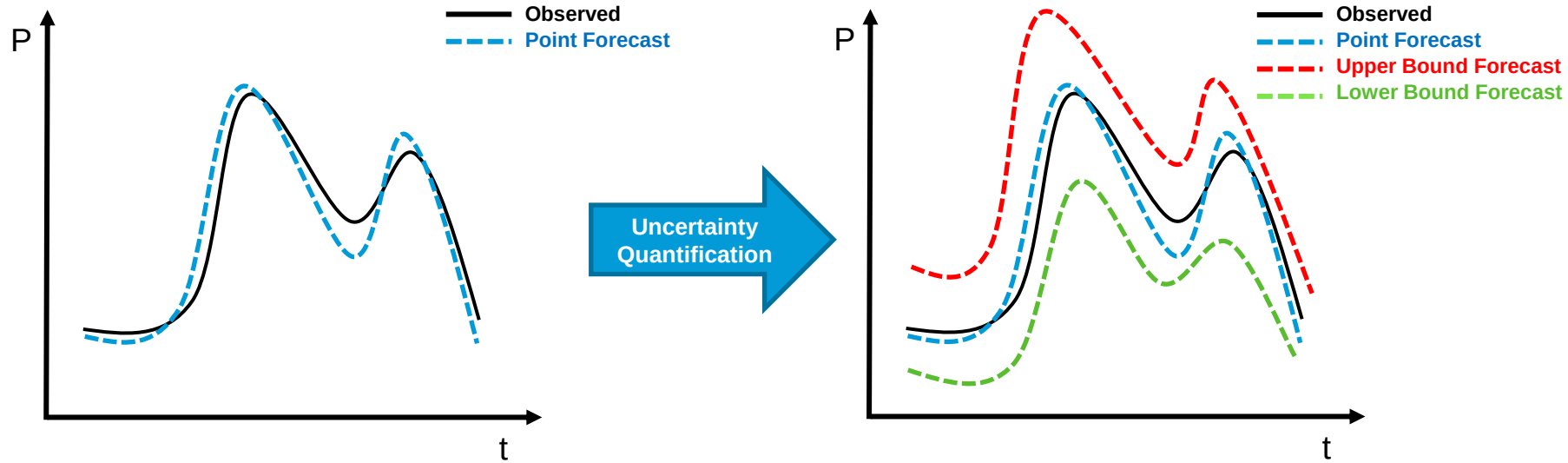
Predicting intervals



Predicting intervals



Predicting intervals

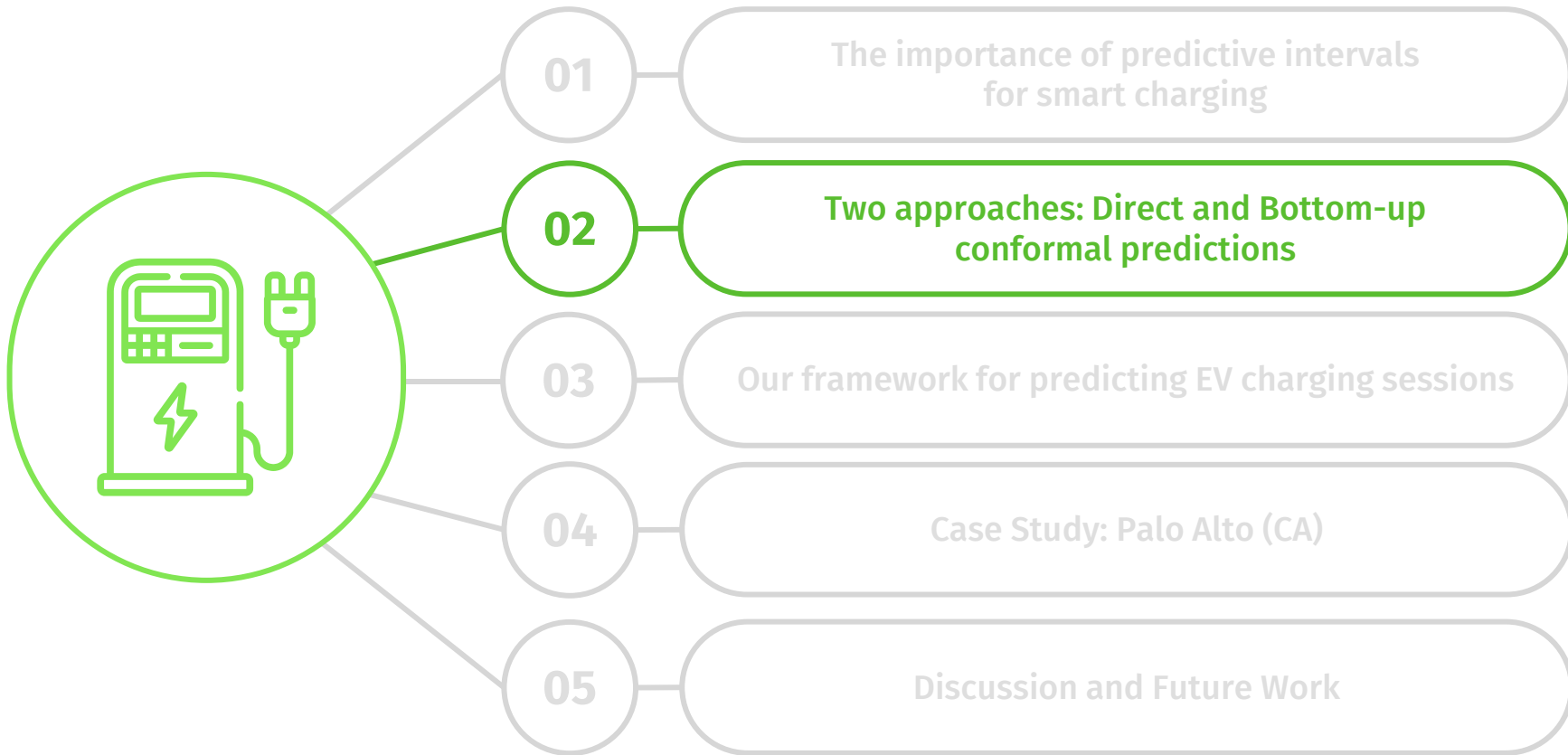


Accurately anticipating incoming EV charging demand is not enough, we need some sense of the uncertainty around the predictions to accommodate for different scenarios.

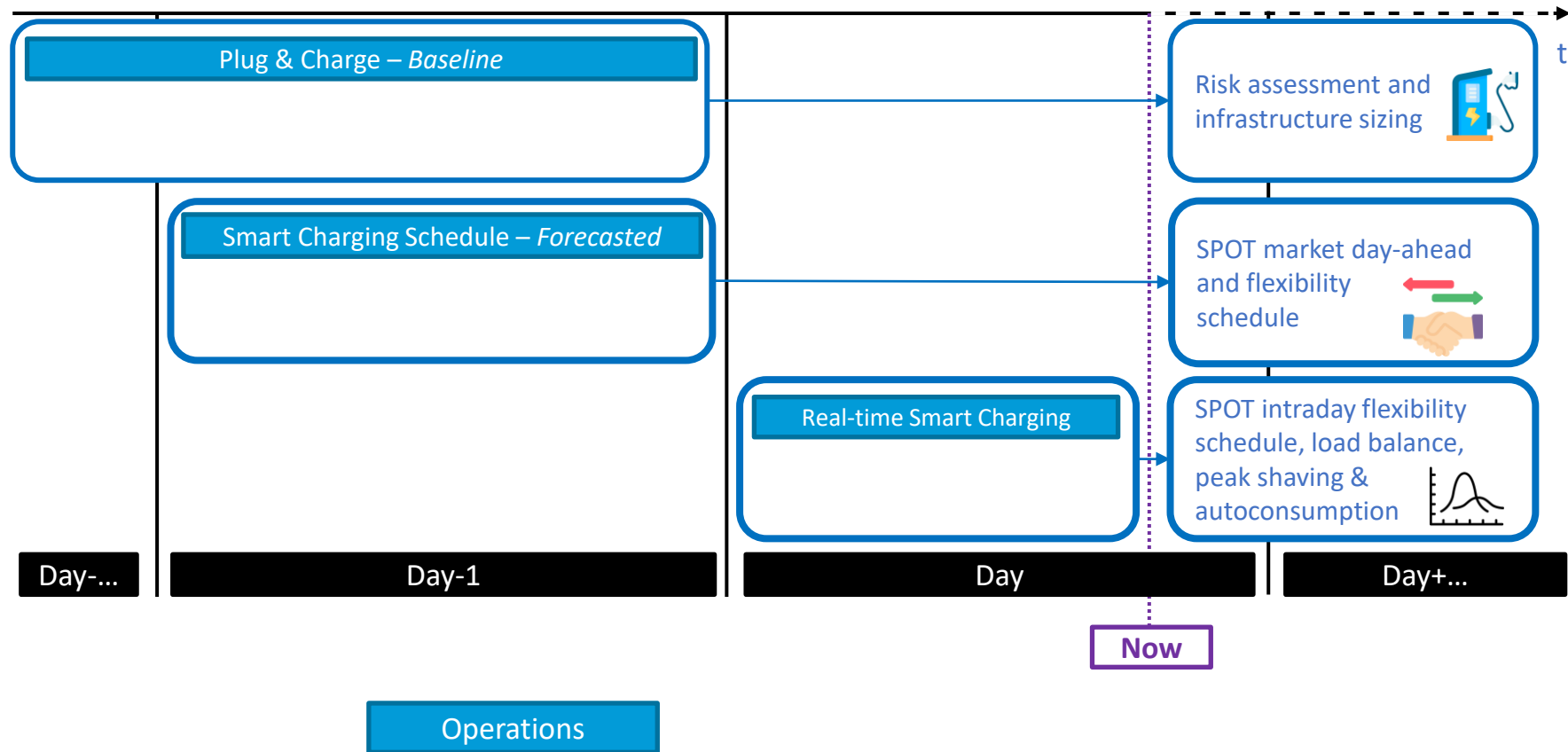
Goal of this work

To assess the quality of various forecasting methods at different time horizons for EV charging demand

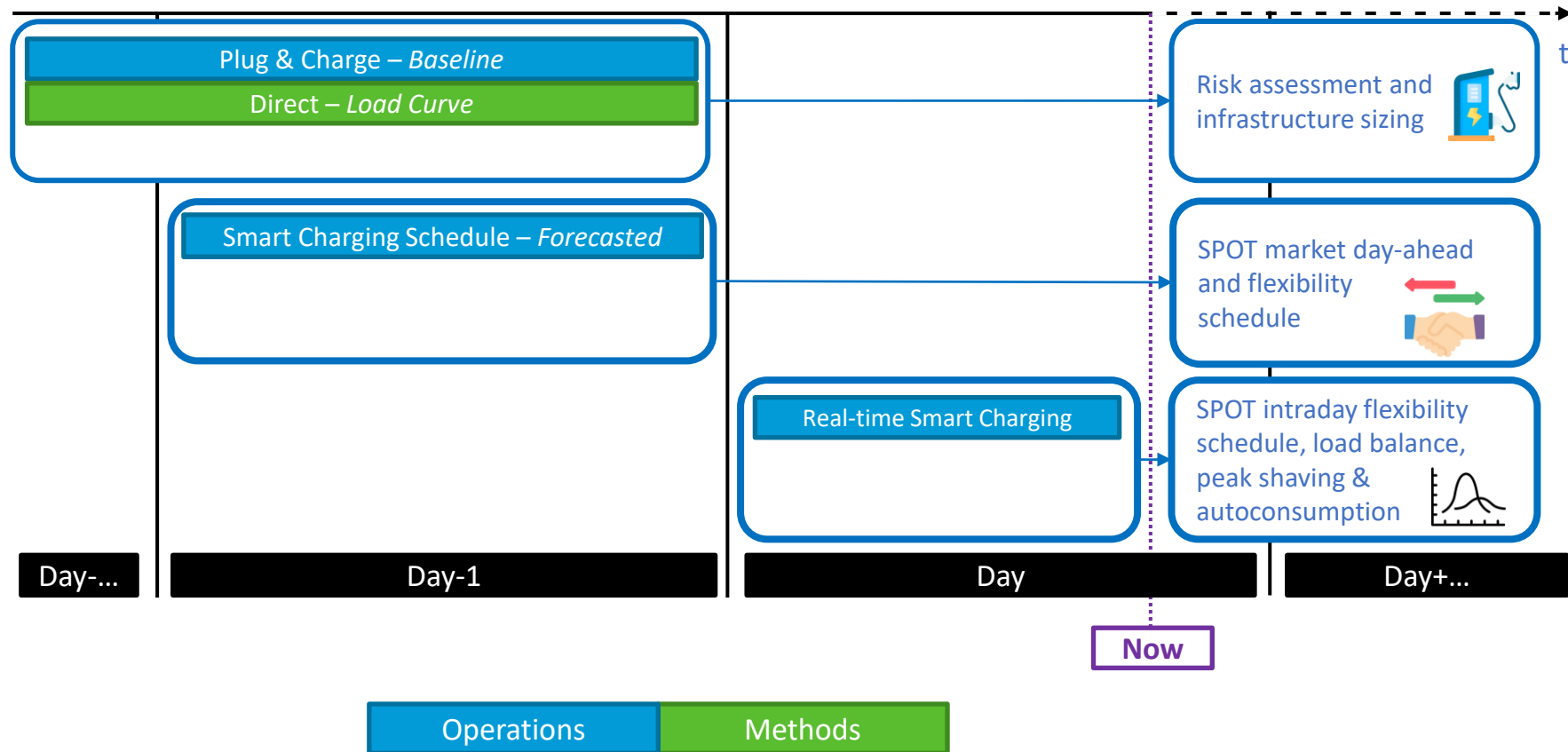
- No inherent bias induced by optimization schedulers in the metrics chosen
- Forecasts are assessed in a plug & charge scenario on real EV transactions on public chargers in Palo Alto (CA)



Operations and methods across the timeline



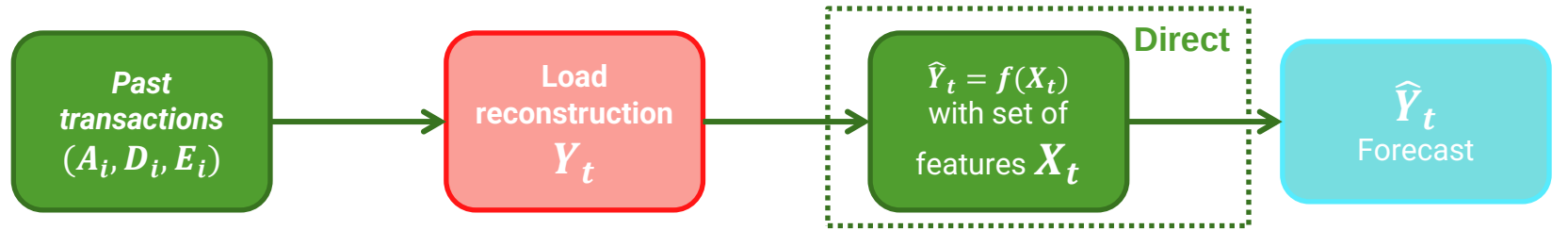
Operations and methods across the timeline



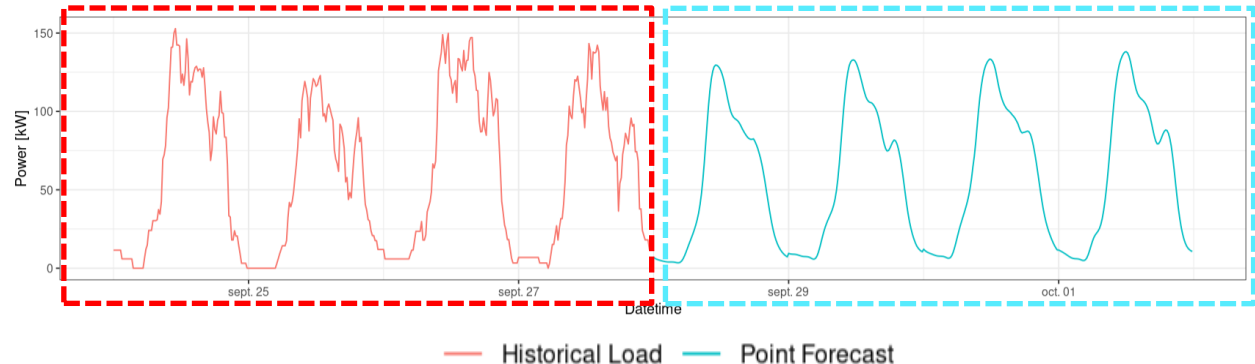
Direct approaches

Interests

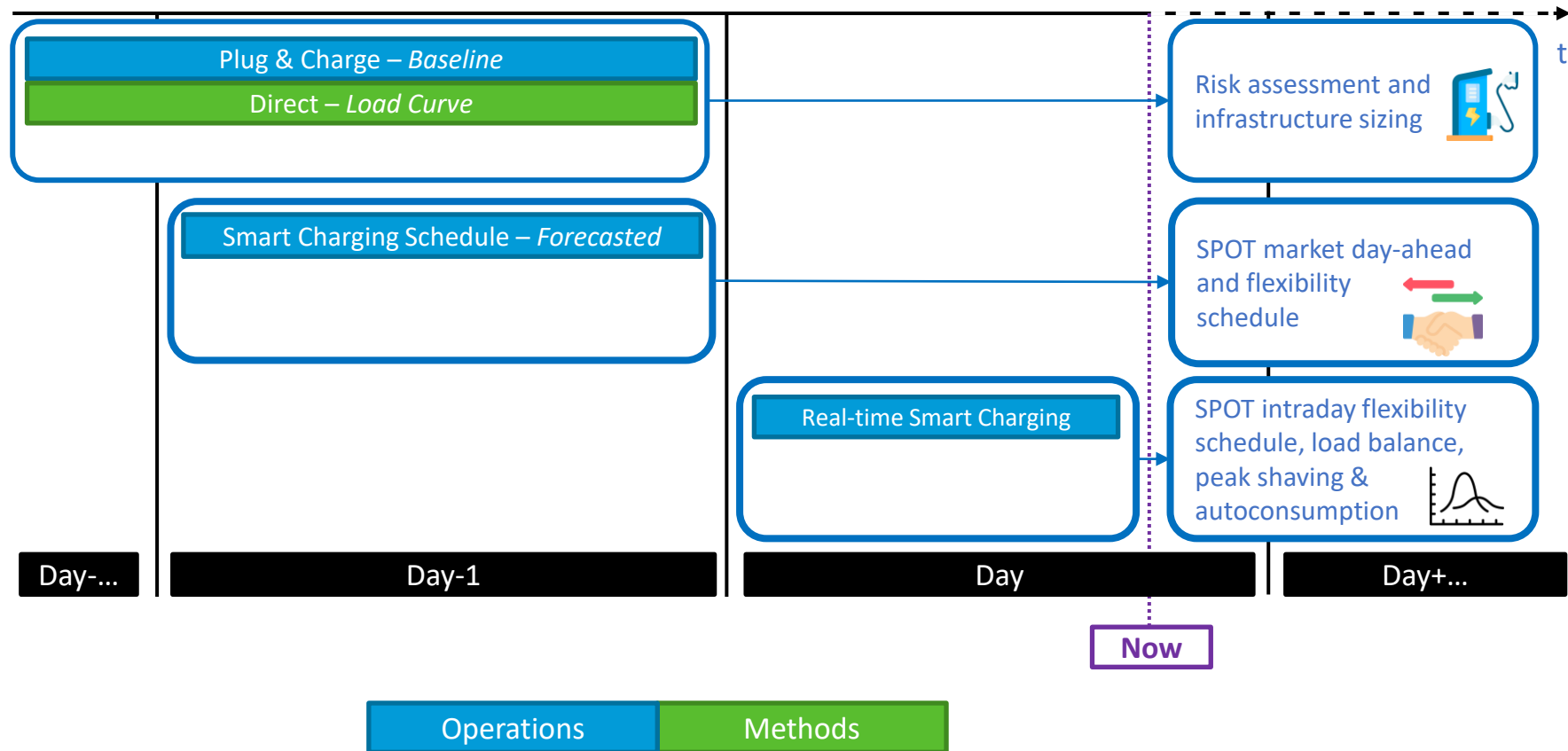
- Plug & Charge load forecasting
- Baseline to which other approaches can be compared



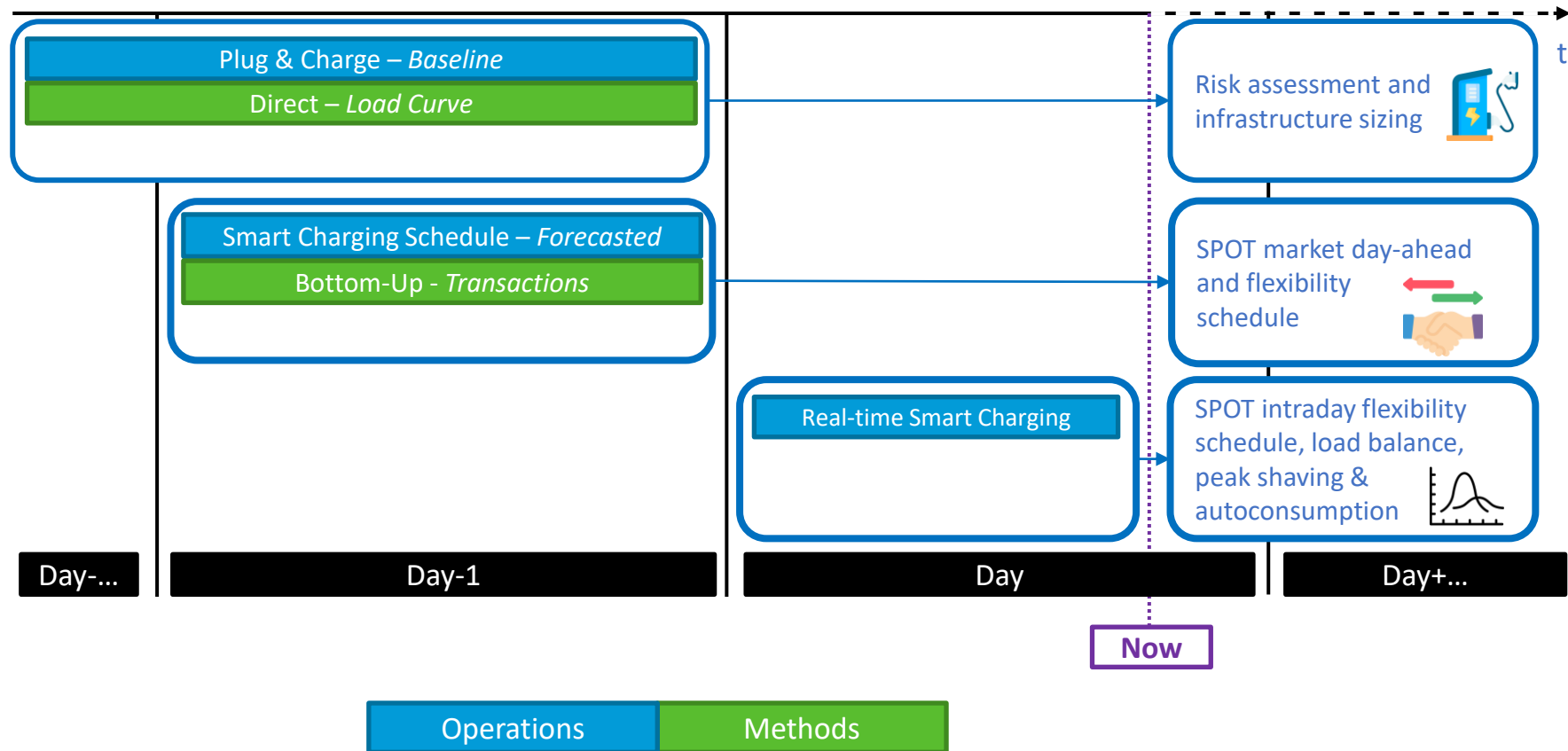
i	A_i	D_i	E_i
1	9h	5h	7kWh
2	12h	2h	3kWh
...	...	...	...
N	17h	1h	1kWh



Operations and methods across the timeline



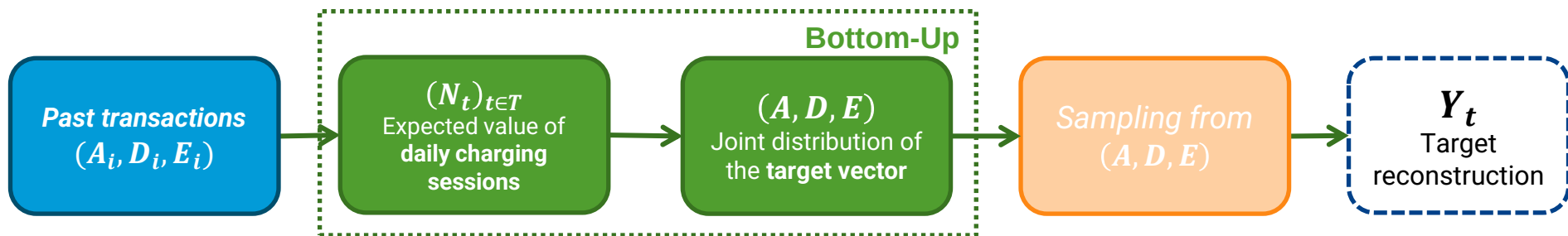
Operations and methods across the timeline



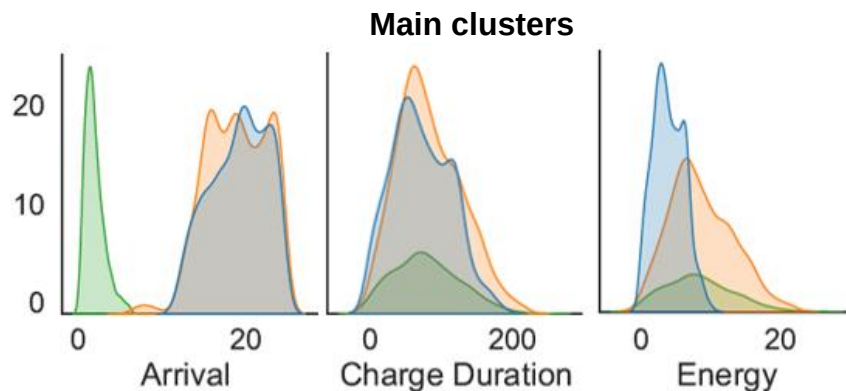
Bottom-Up approaches

Interests

- **Finer granularity of forecasting (at the transaction level)**
- **Allows smart charging applications**
 - More leverage to optimise at the individual curve level
 - Optimisation on intra-day markets
 - Self-consumption and V2H applications

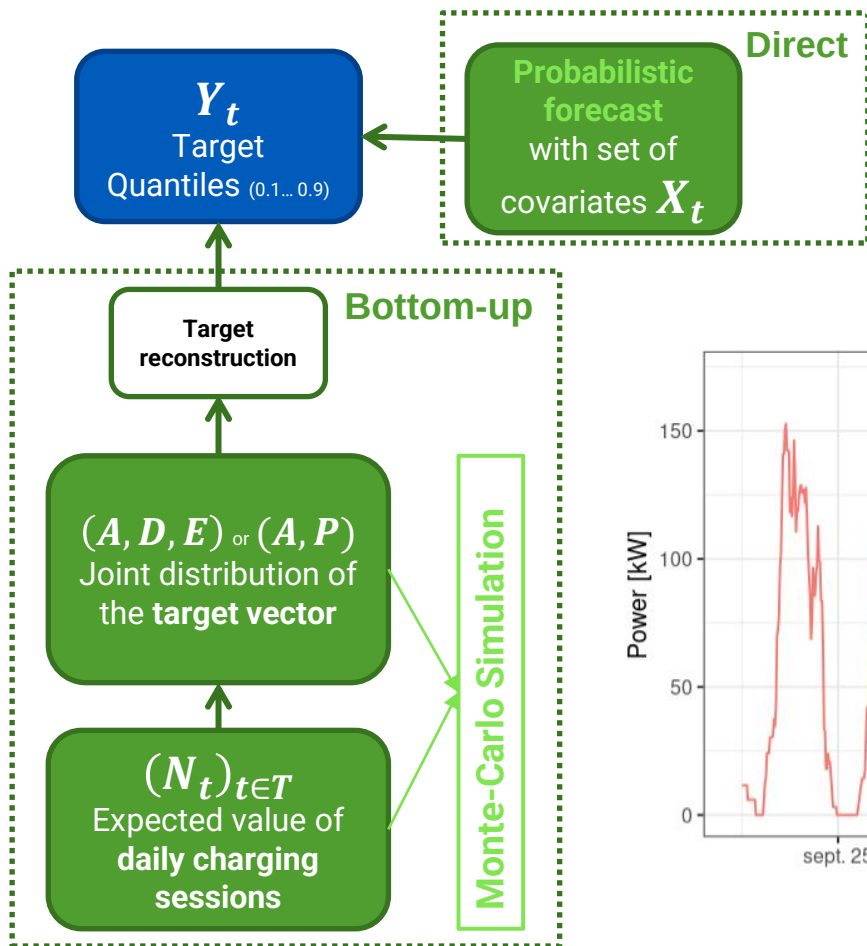


i	A_i	D_i	E_i
1	9h	5h	7kWh
2	12h	2h	3kWh
...	...	...	...
N	17h	1h	1kWh



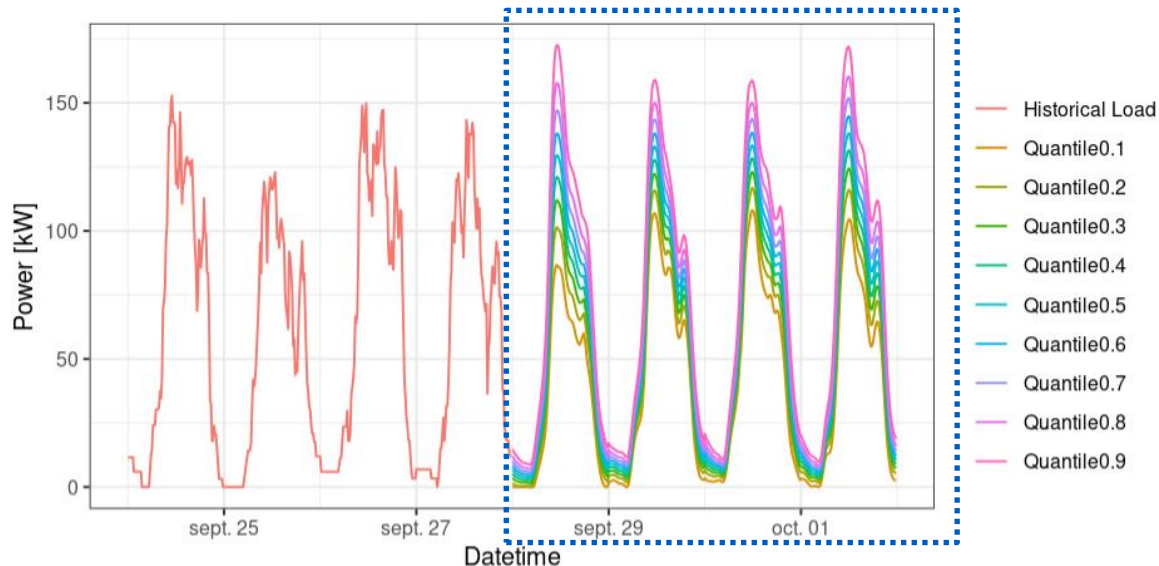
i	A_i	D_i	E_i
1	8h	4h	9kWh
2	11h	1h	1kWh
...	...	...	...
$\hat{N}$	13h	2h	1kWh

From point forecasts to probabilistic forecasts

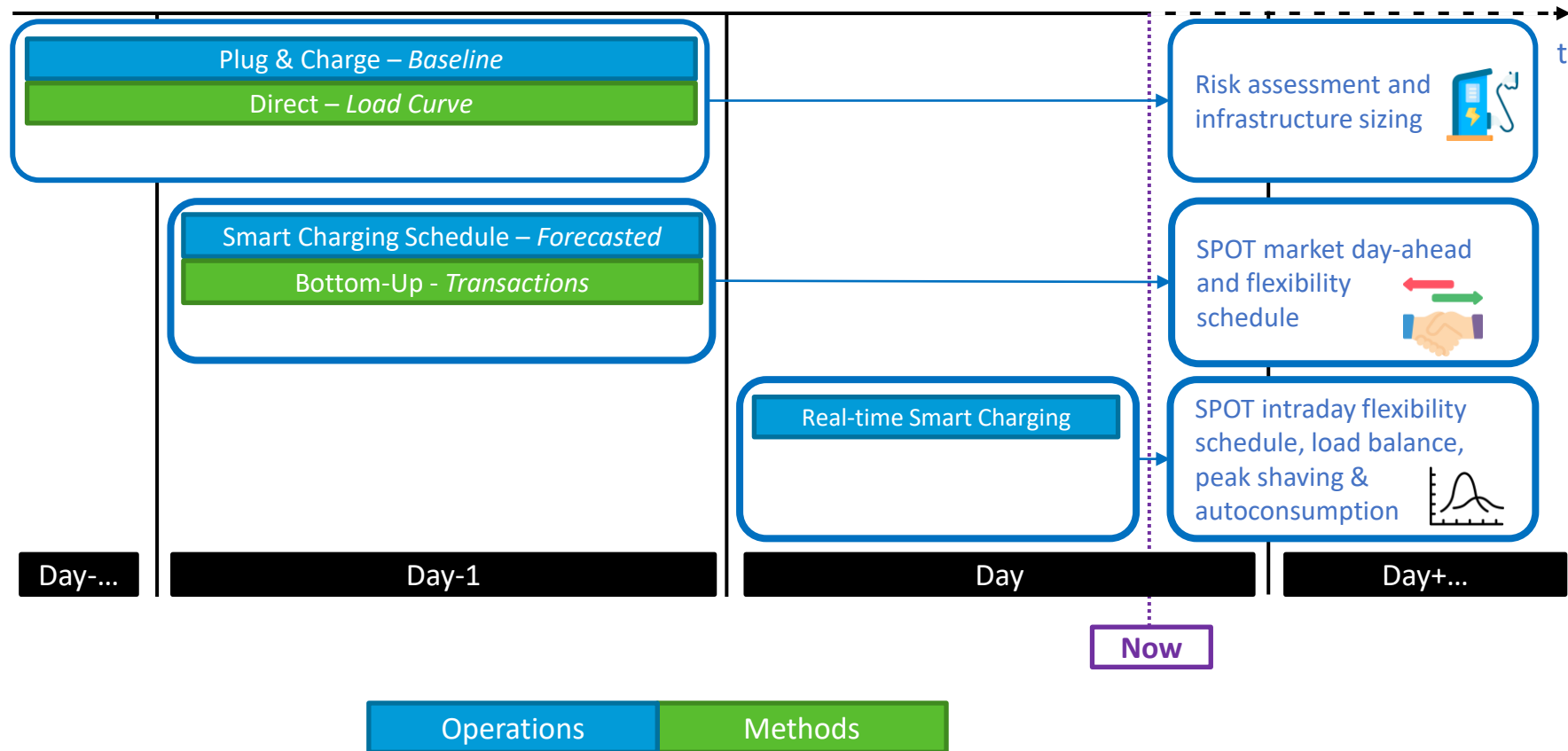


Interests

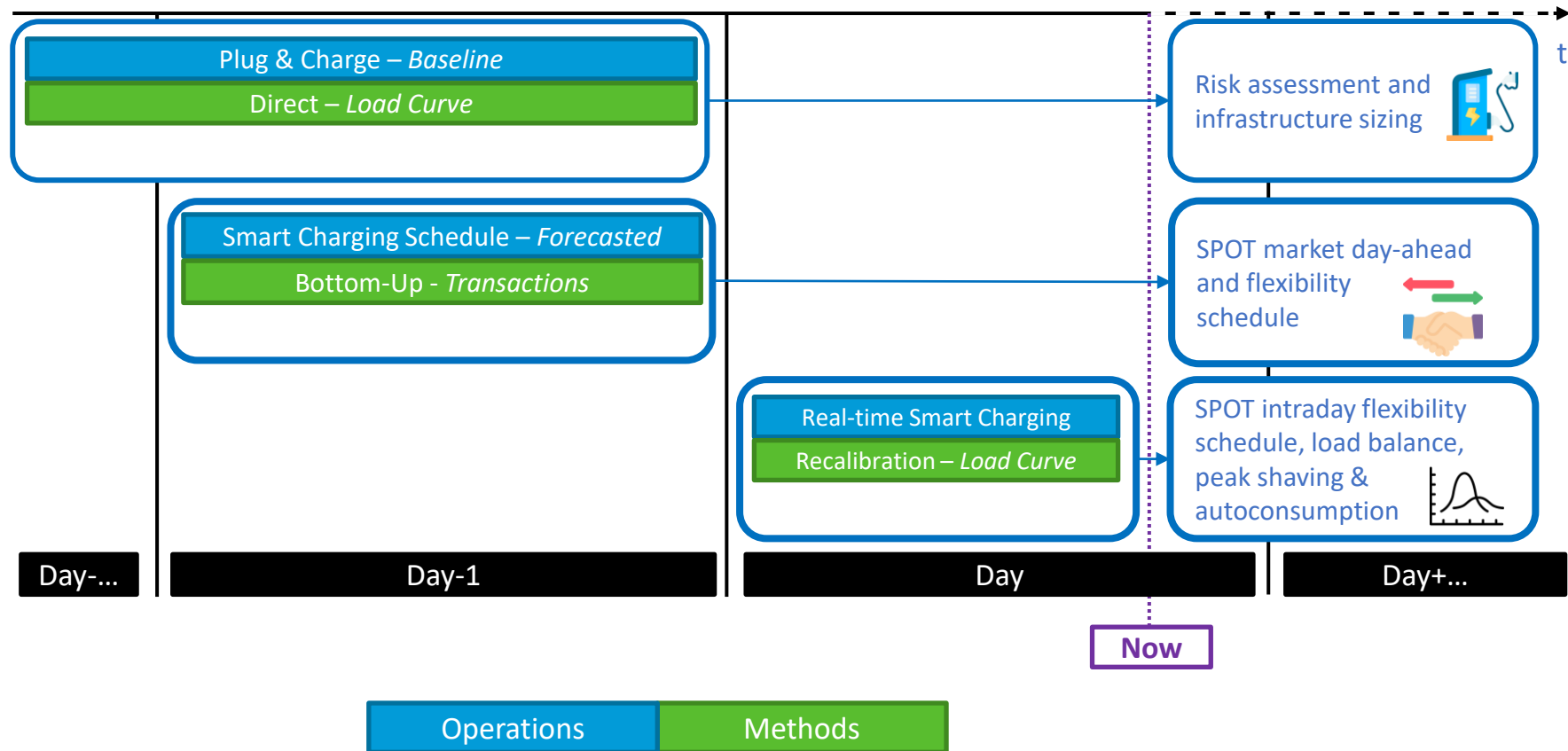
- Need to quantify the uncertainty around these forecasts to assess the risk around the smart charging operations
- The optimisation is different if there is an important uncertainty around the expected load (higher energy, shorter durations etc.)



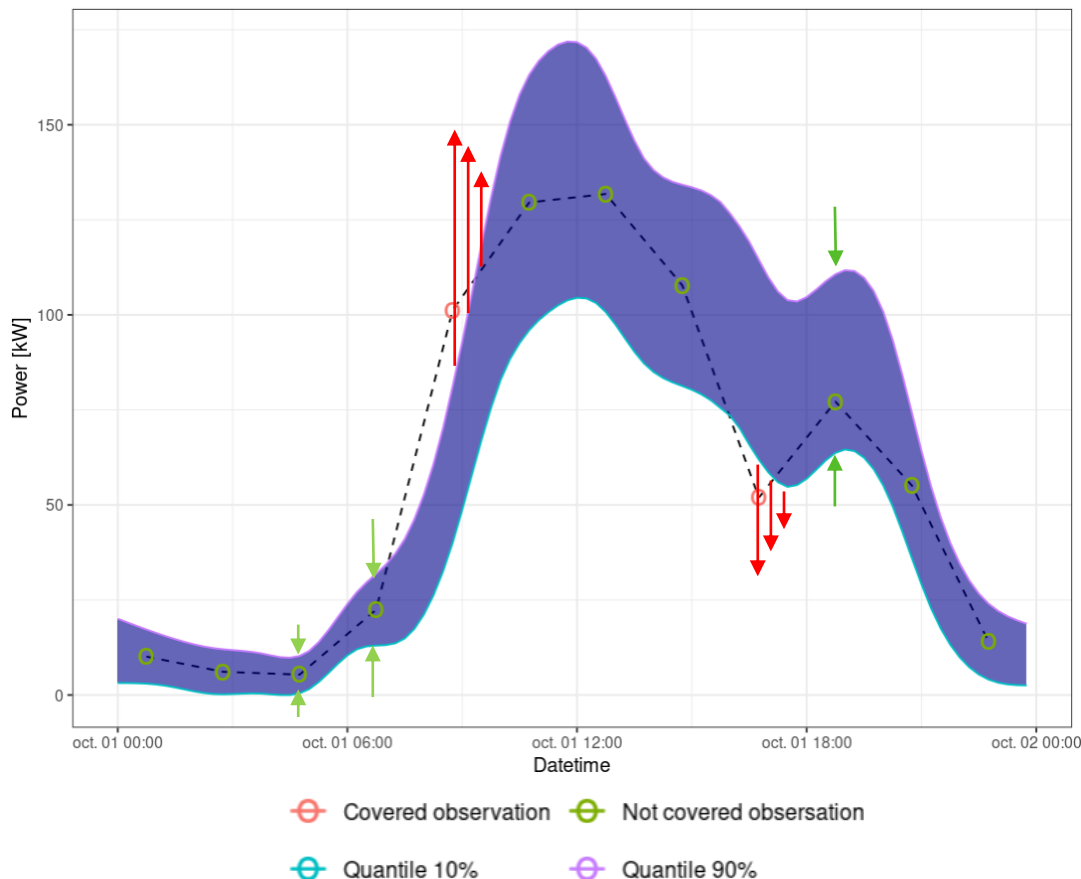
Operations and methods across the timeline



Operations and methods across the timeline



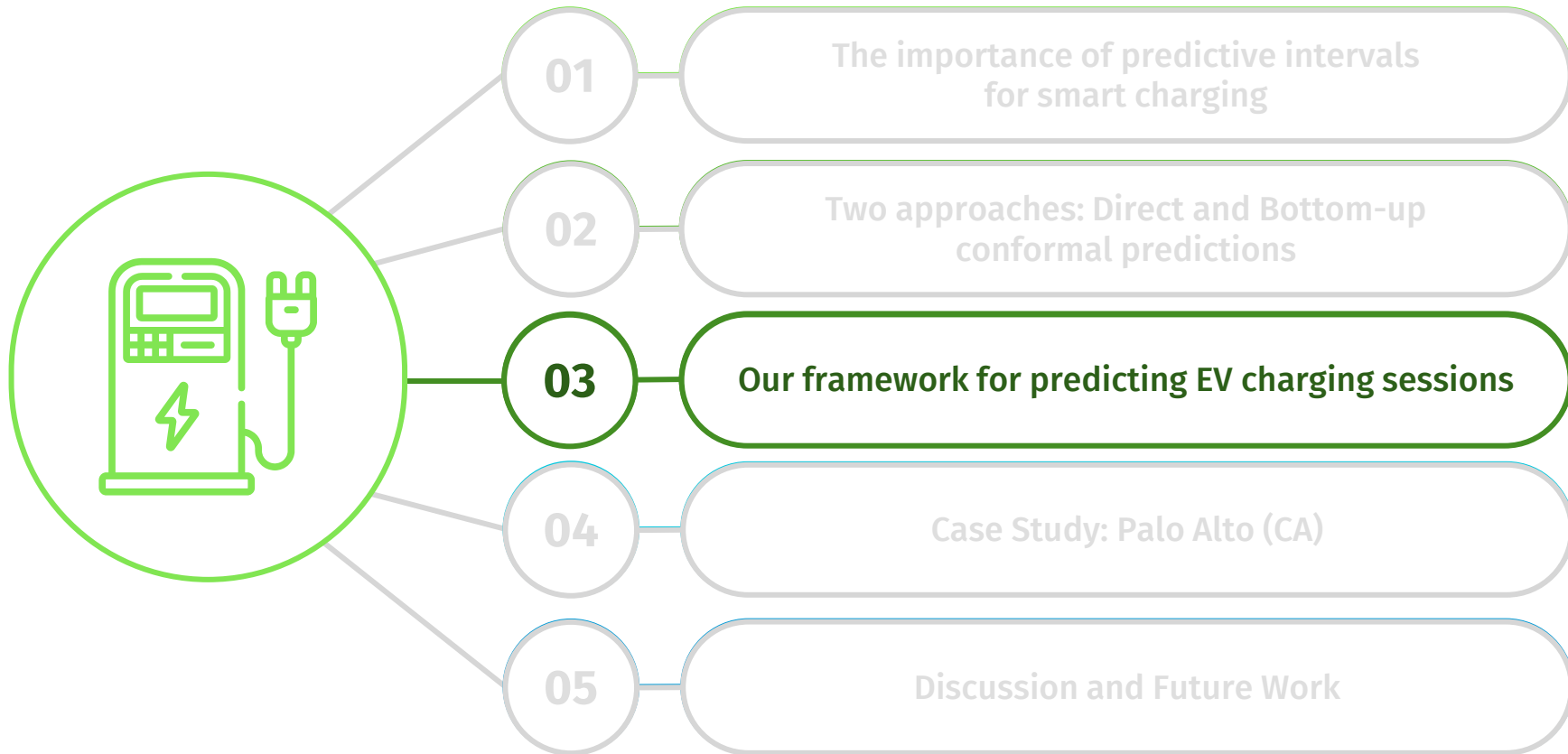
Recalibration



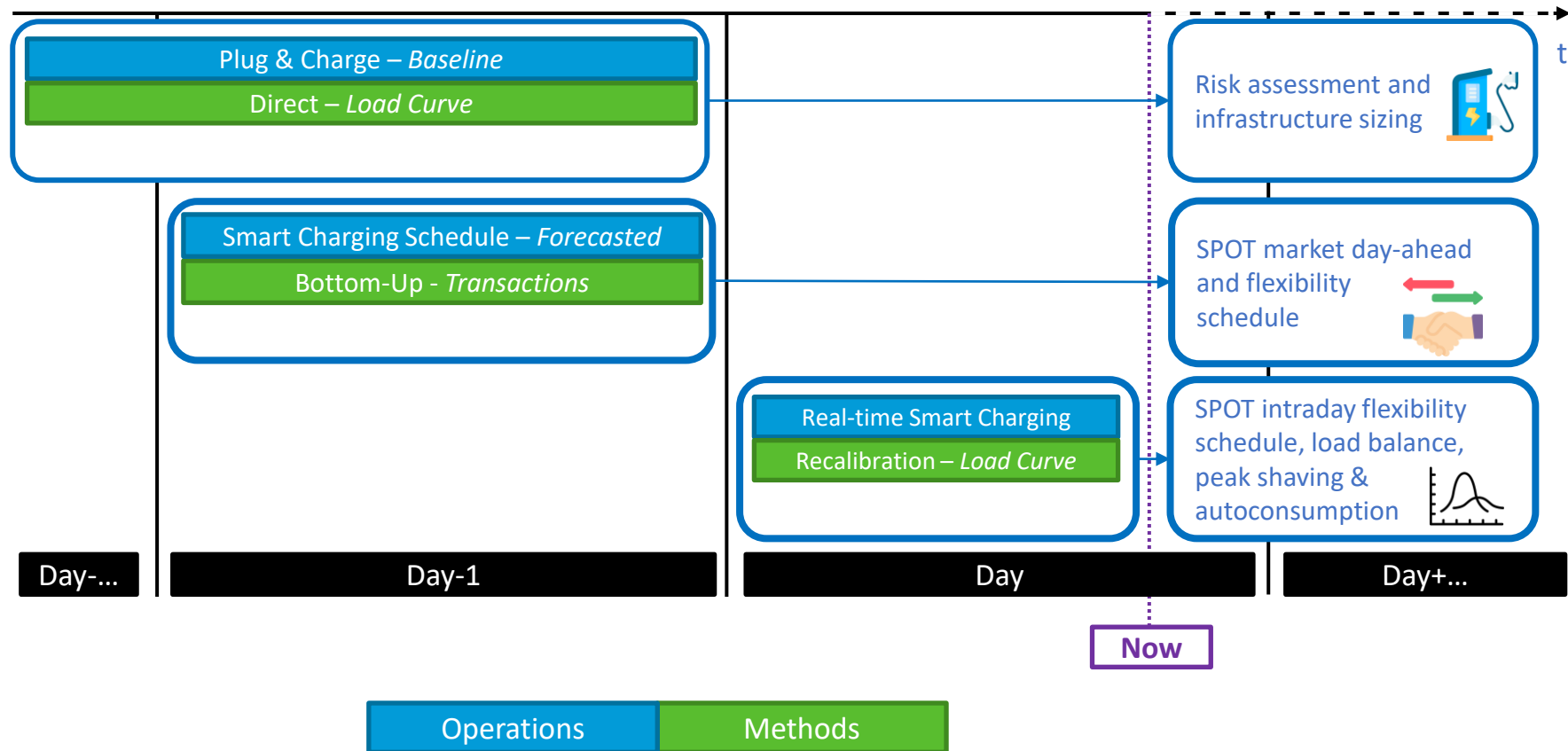
Interests

- A lot of uncertainty around the forecasts especially at the individual transaction level
- Use the latest observations available to correct the forecasts and have a better view of the user's behavior before applying smart charging algorithms

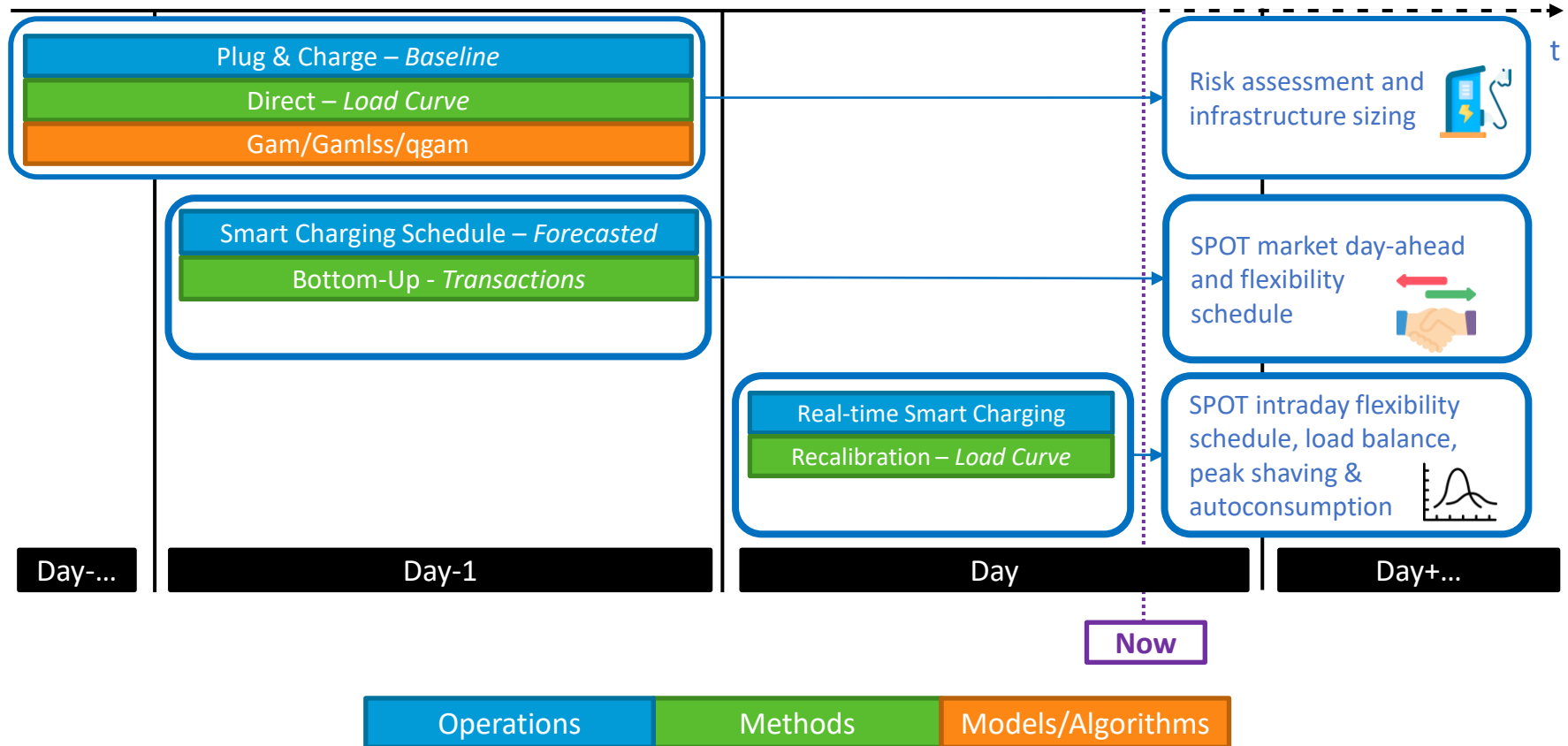
$$Coverage = \frac{1}{T} \sum_{t=1}^T 1(\hat{y}_{10\%}(t) < y_{obs}(t) < \hat{y}_{90\%}(t))$$



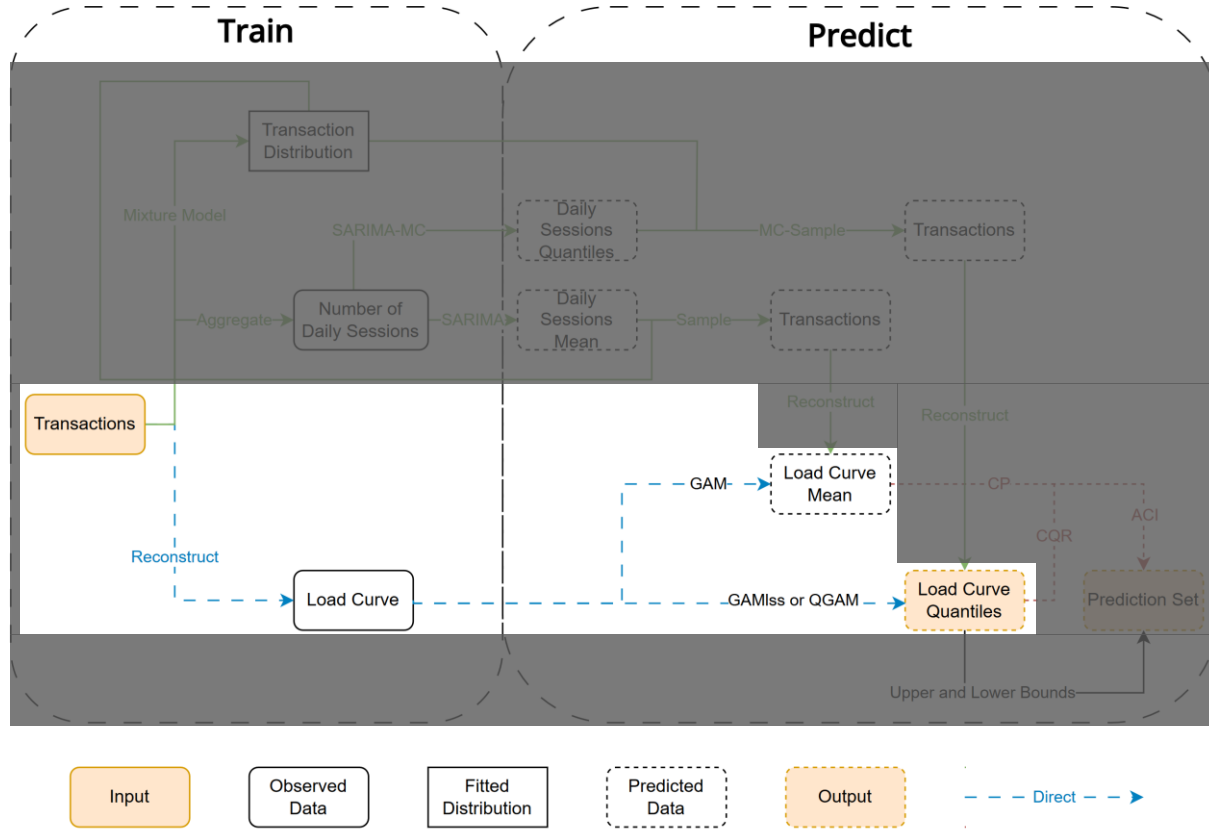
Operations, methods and models across the timeline



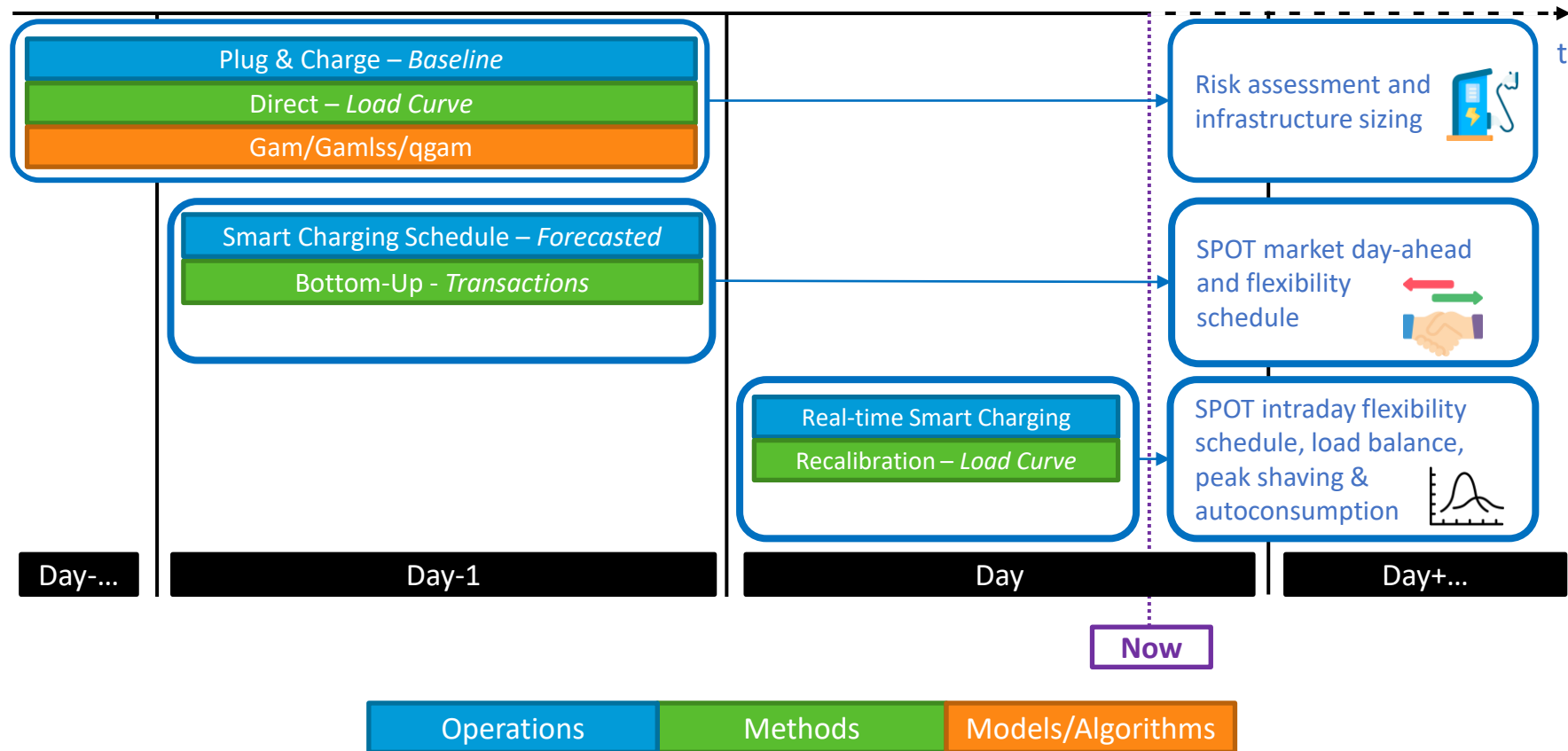
Operations, methods and models across the timeline



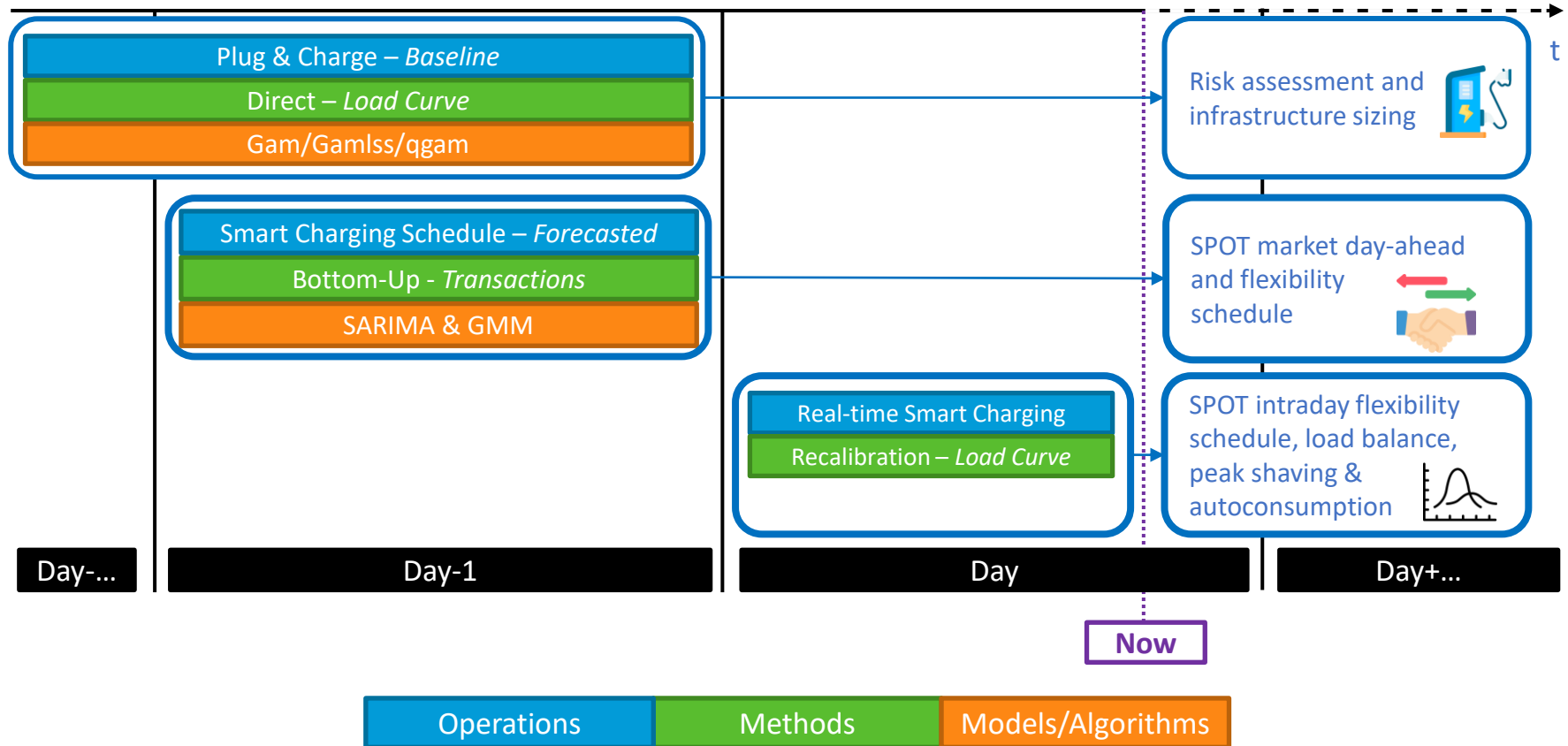
Prediction Framework



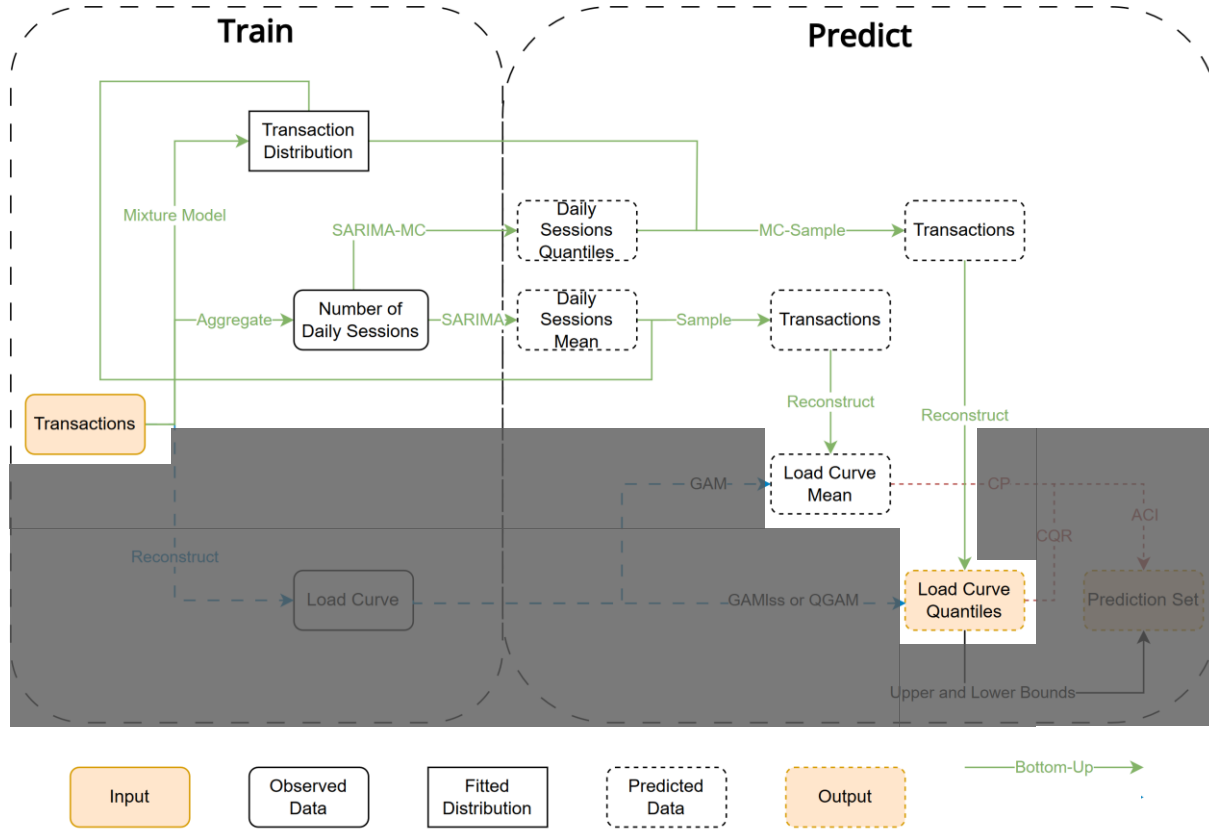
Operations, methods and models across the timeline



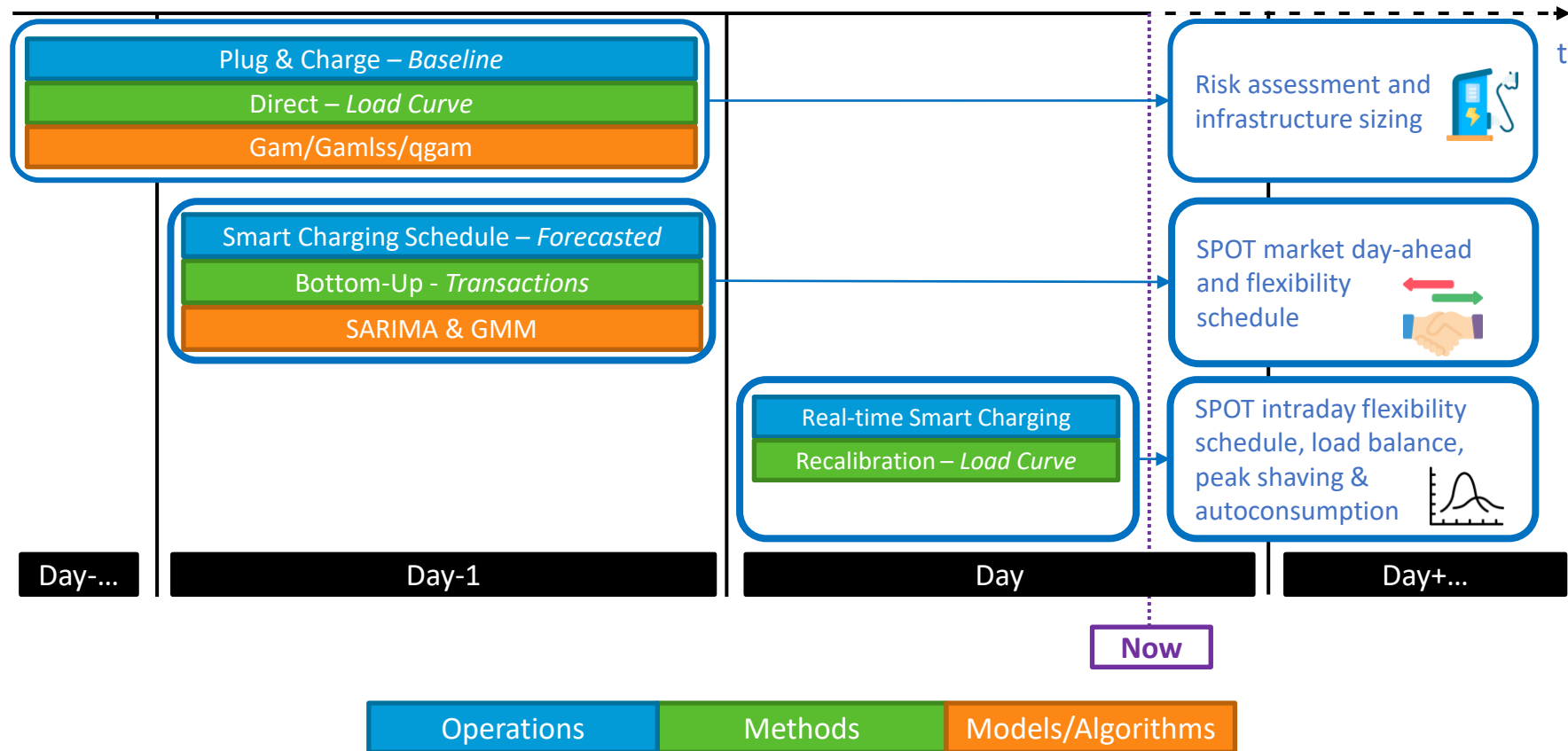
Operations, methods and models across the timeline



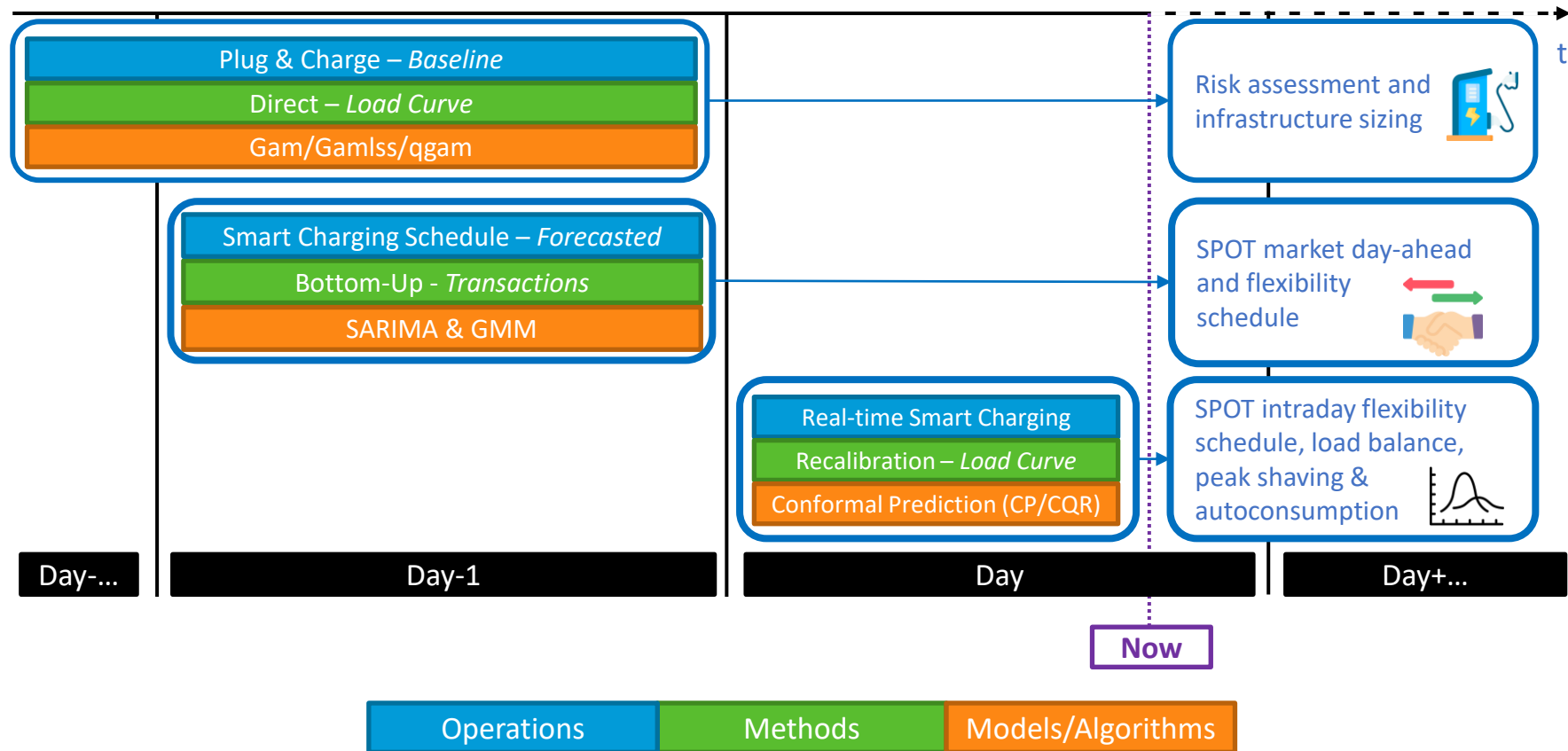
Prediction Framework



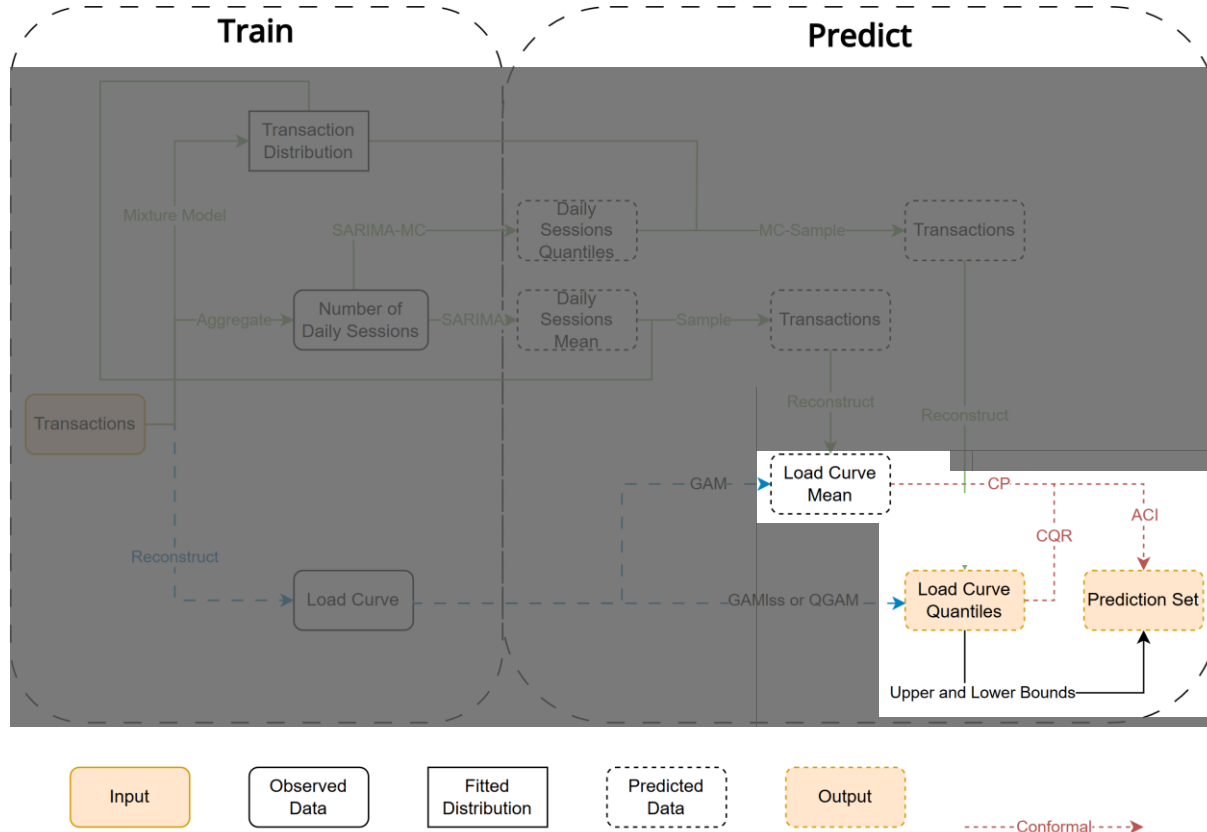
Operations, methods and models across the timeline

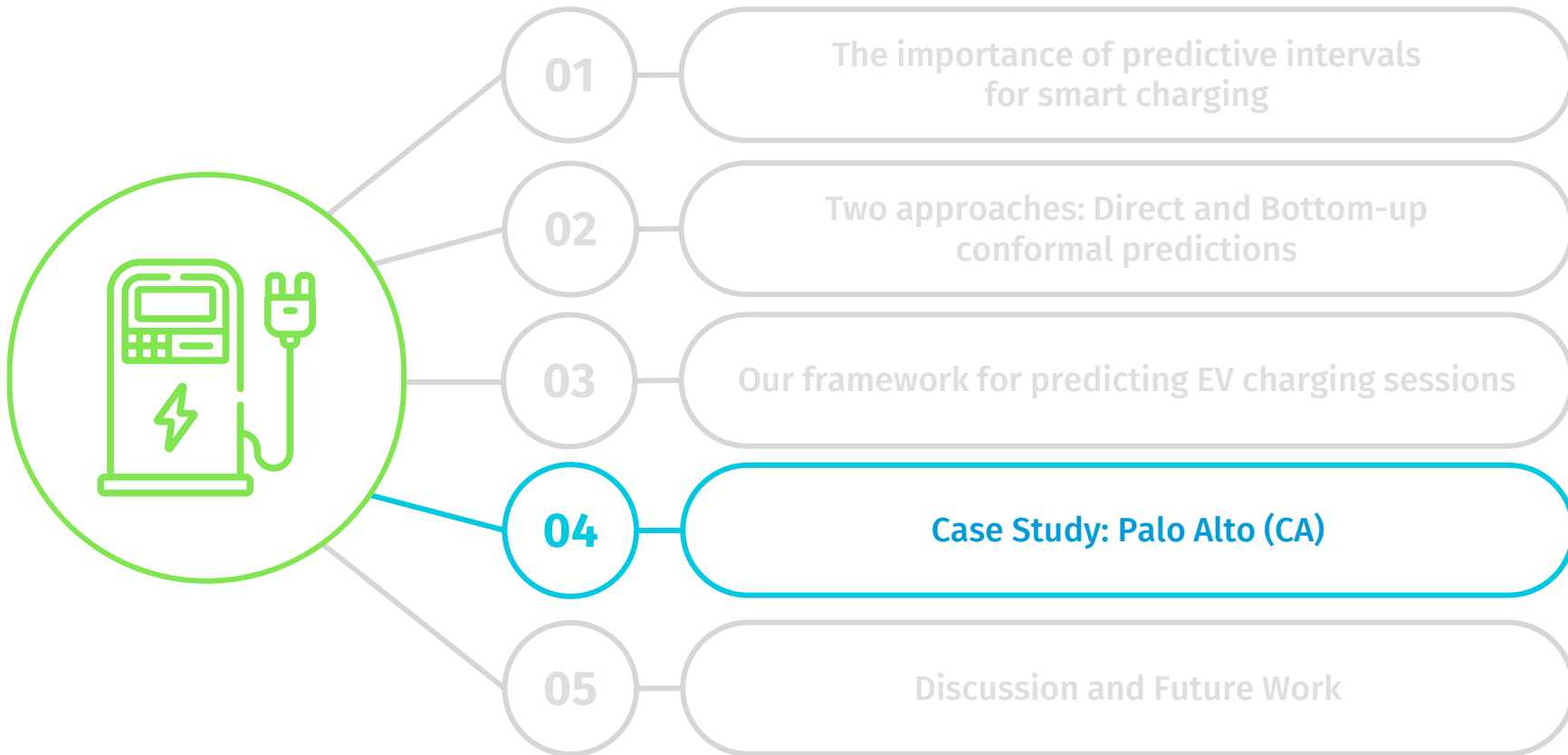


Operations, methods and models across the timeline



Prediction Framework





Palo Alto Dataset

Public Charge

Boulder (USA)

Dundee (UK)

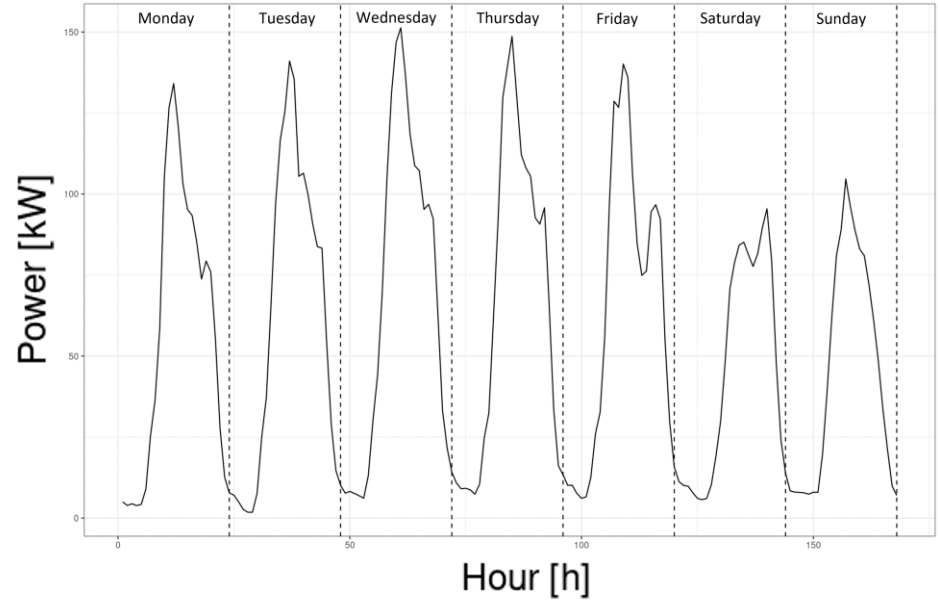
Palo Alto (USA)

Paris (FRA)

Perth (UK)



Open Data on public charge available on GitHub:

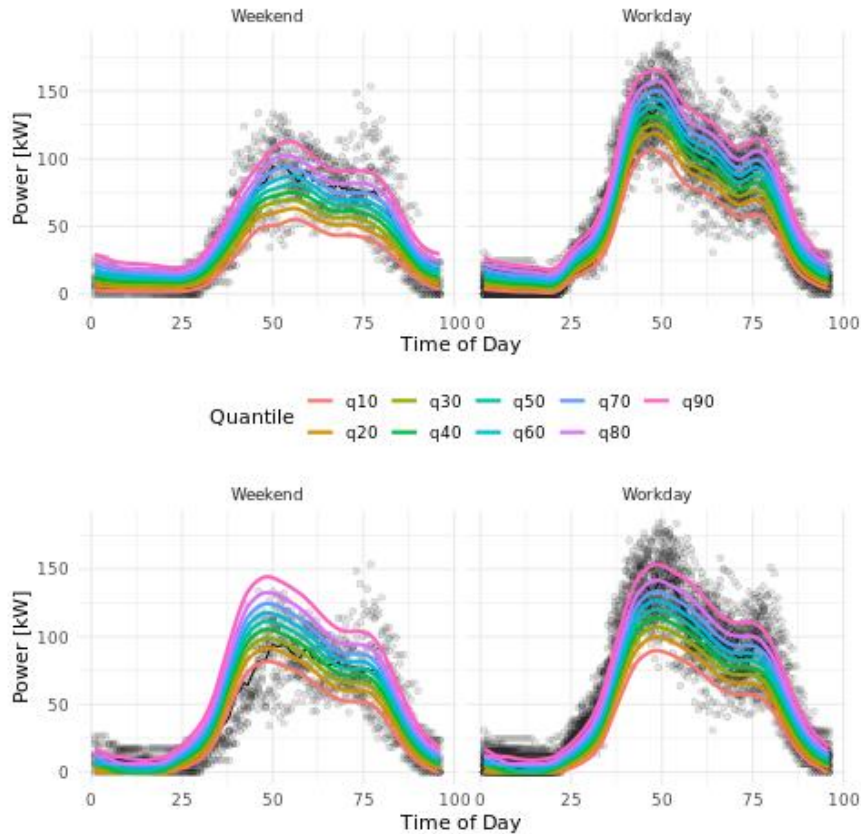


Load profiles for a typical week in Palo Alto (CA)

Uncertainty quantification from Direct and Bottom-Up

Results

- Direct Approaches give the best results in terms of day ahead forecasts for the plug & charge use case
- Bottom-Up gives good results on this task even though it is a more challenging problem
- Among the probabilistic methods, qgam is the best one over almost every quantile, both in accuracy and in sharpness



Averaged quantile forecasts (distinguished by type of day) 40

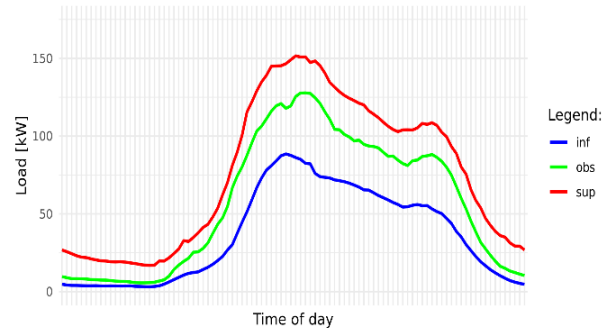
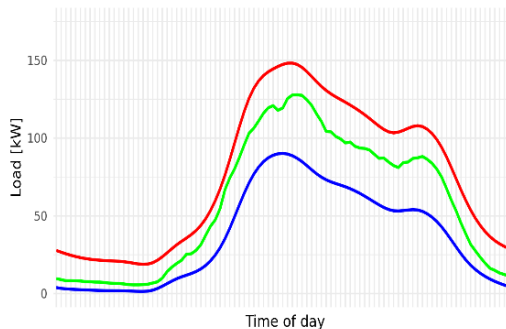
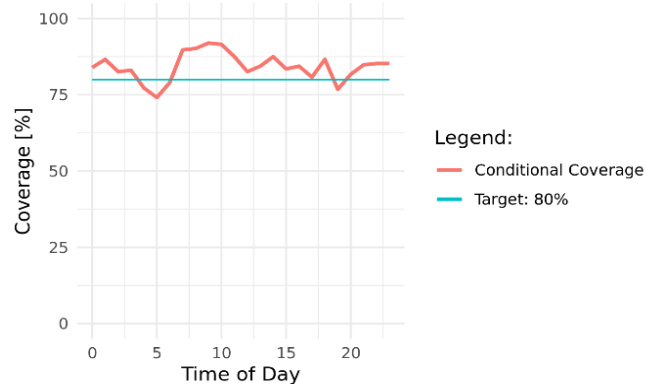
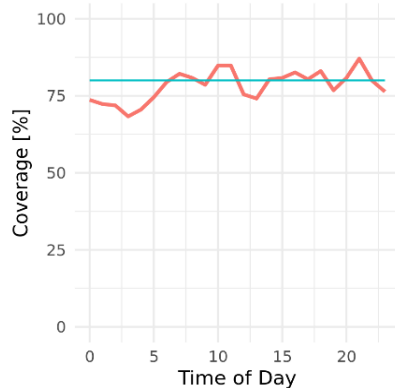
Improvement with conformal prediction

From static to adaptive forecasts

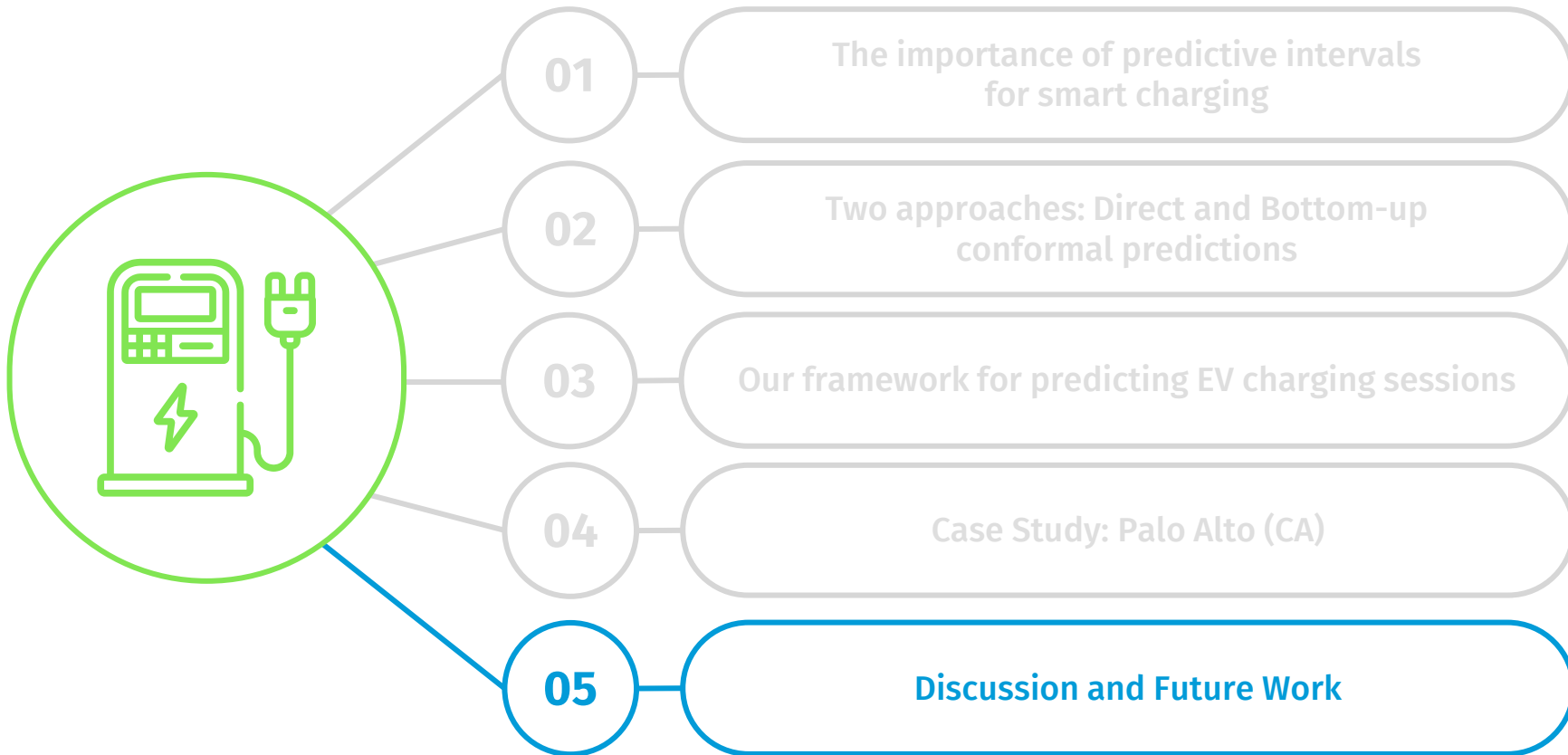
- Both Direct and Bottom-Up methods cannot adapt to behaviour changes
- This can lead to miscalibrated uncertainty in the real-time setting
- To address this issue, we use Adaptive Conformal Inference methods

Results

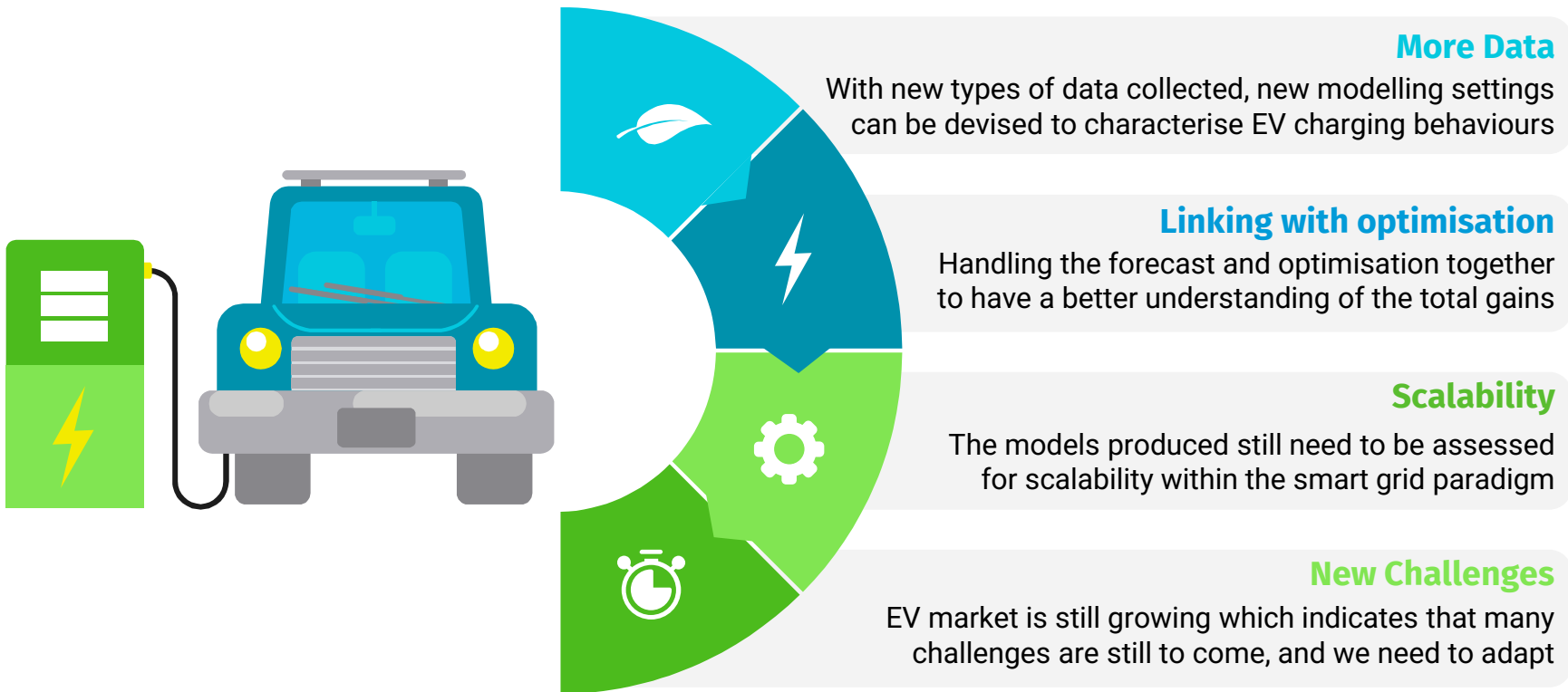
- Higher Empirical Coverage while maintaining conditional Coverage
- Equivalent interval width on daily profile, even sharper during night



Conditional Coverage and daily profiles for qgam and CQR qgam



Perspectives



Thank you for your attention



Yvenn Amara

EDF R&D and Univ. Paris-Saclay

Ass. Professor and
Research Engineer



Bachir Hamrouche

EDF R&D

Research Engineer



Guillaume Principato

Université Paris-Saclay

PhD Candidate



Yannig Goude

EDF R&D and Univ. Paris-Saclay

Ass. Professor and
Senior Research Engineer

Relevant Publications

1. Amara-Ouali, Y., Goude, Y., Massart, P., Poggi, J. M., & Yan, H. (2021). A review of electric vehicle load open data and models. *Energies*
2. Amara-Ouali, Y., Goude, Y., Hamrouche, B., & Bishara, M. (2022). A benchmark of electric vehicle load and occupancy models for day-ahead forecasting on open charging session data. In *Proceedings of the Thirteenth ACM International Conference on Future Energy Systems*
3. Amara-Ouali, Y., Goude, Y., & Poggi, J. M. (2023). Modelling the Intensity of Electric Vehicle Arrivals at Charging Points. *Transportation Research Procedia*